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Adaptive and highly accurate numerical treatment for a gradient-enhanced brittle damage model

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Summary

We present an adaptive numerical treatment for a recently published model for brittle damage with gradient enhancement. We employ adaptive strategies both in time and space, yielding detailed investigations of the convergence behavior: the damage distribution can be resolved with very high accuracy, turning the damage behavior into cracks known from fracture mechanics. Although the localization is extreme, mesh-independent results are obtained thanks to the spatial adaptivity and allow for a detailed investigation of the influence of the regularization parameter.

KEYWORDS

adaptive finite element method, brittle damage, gradient-enhanced regularization, Laplace operator, snapback

1 | INTRODUCTION

Material and structural failure can be modeled in various ways. For instance, fracture mechanics allows for a detailed analysis of the stress field at a crack tip.¹⁻³ The eigenerosion method, see References 4 and 5, might be interpreted as a (numerical) extension of fracture mechanics. Phase-field models are also feasible for the description of failure.⁶⁻⁹ Finally, damage models might be used since they use the simplest possible assumption that, after reaching some critical threshold value, stiffness decreases with increasing loads. This might be formulated in terms of isotropic, cf. References 10 and 11, and anisotropic damage, cf. References 12 and 13. Due to the simple assumptions for damage models, they have been used for modeling failure events at various scales, see, eg, References 11,14-17, including structural effects such as the snapback phenomenon.^{16,18,19}

However, damage models inherently constitute an ill-posed mathematical problem: due to their softening character, a loss of ellipticity of the resultant system of algebraic equations yields a strong mesh dependence when evaluated within a finite element routine, see, eg, References 19 and 20. Consequently, methods for regularization have to be employed, of which three different classes exist: (i) viscous regularization, (ii) integral formulations, and (iii) gradient enhancement.

Viscous regularization, such as in References 21-26, is the most direct approach for solving the issue of mesh-dependent finite element results. The main advantage of this strategy is the easy mathematical formulation, which allows for a numerical implementation that can be carried out at the material point level. To be more precise, the usage of a material law subroutine is sufficient. Unfortunately, there exists a remarkable drawback that prevents viscous regularization to become the dominant strategy chosen: the results explicitly depend on the loading velocity, which might contradict experimental observations, particularly when brittle material failure is investigated. Furthermore, model calibration is

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cumbersome since usually viscosities must be used to obtain mesh independent results that are unphysically high, such that they do not constitute as a material parameter that is independent of the respective boundary value problem. In contrast, loading velocity in combination with the damage viscosity has to be calibrated for each boundary value problem on an individual basis.

Integral formulations and gradient enhancement, which show some mathematical similarities to phase field approaches, share the characteristic that they usually demand a more sophisticated strategy for numerical implementation: novel finite elements have to be programmed in which the damage evolution is averaged over same neighborhood of elements (for integral formulations), or in which additional nodal unknowns enter the problem (for gradient enhancement). While these strategies consume more computational effort than viscous regularization, the huge benefit is that such models yield mesh-independent finite element results for arbitrary loading velocities and can also be used for modeling brittle damage. Examples are given in References 3, 19, 20, 27, and 28.

Additional model parameters, such as the damage viscosity for viscous regularization, are also present both for integral formulations and gradient enhancement which play here a less significant role due to physical reasons but are a mathematical necessity. For integral formulations, the radius of the circle (sphere) over which the averaging is performed has to be defined. Prominent examples are the so-called filter techniques commonly used for topology optimization problems (which might be interpreted as “inverse damage” problems).^{29,30} The regularization parameter used in strategies based on gradient enhancement has a similar effect. Here, the parameter directly controls the width of the damage zone and also the transition from the undamaged to the damaged area (large values yield thick damage zones and a very smooth transition, whereas small values yield crack-like damaged zones with very sharp interfaces between damaged and undamaged material). A physical interpretation of the regularization parameter is that it reflects the interface energy and that it is related to the “internal length” given by the crack width, for example, Reference 27.

Although gradient enhancement yields mesh-independent results, the impact of the chosen penalization parameter is huge: not only the shape of the damage zone is controlled but also the maximum load before material failure strongly depends on the particular value for the parameter. In order to resolve a distinct damage zone in greater detail, strategies for mesh adaptivity can be employed.³¹ However, in contrast to the accepted mesh dependence in Reference 31 due to missing regularization, which was assumed to be adjustable to experiments, a rigorous consideration of gradient enhancement should be maintained to still deal with a well-posed mathematical problem.

The motivation for this article is to investigate the capabilities of the gradient-enhanced material model for brittle damage in Reference 32, where the neighbored element method (NEM) was used. It showed the huge advantage that each global load step could be computed with very similar numerical effort, that is, computation time, as a load step for a purely elastic material behavior. In this article, we improve the NEM such that is in particular capable of handling adaptive finite element meshes and present a detailed and advanced study on the numerical behavior of this method for adaptive treatment of the meshing and load stepping at runtime.

The article is structured as follows: we begin with a short presentation of the model in Reference 32 for convenience. Afterward, we present the novel numerical treatment and its implementation, with special emphasis on both the finite element implementation of the model itself and the adaptive strategies applied. Then, we perform a detailed numerical study that investigates all important effects concerning convergence behavior and the dependence on the material and model parameters.

2 | GRADIENT-ENHANCED MODEL FOR BRITTLE DAMAGE

The model is derived employing Hamilton's principle, which belongs to the variational methods in material modeling and is given by

$$\delta \mathcal{H} = \delta \mathcal{G} - \delta \mathcal{W}_{\text{diss}} = 0 \quad \forall \delta \mathbf{u}, \delta d, \quad (1)$$

with the Gibbs energy $\mathcal{G}$, the virtual work due to nonconservative forces $\delta \mathcal{W}_{\text{diss}}$, the displacement $\mathbf{u}$, and the damage parameter d . The Gibbs energy is given by

$$\mathcal{G} := \int_{\Omega} \Psi(\epsilon, d) dV - \int_{\Omega} \mathbf{b} \cdot \mathbf{u} dV - \int_{\Gamma_{\sigma}} \mathbf{t} \cdot \mathbf{u} dS, \quad (2)$$

with the Helmholtz free energy Ψ and the virtual work due to nonconservative forces by

$$\delta \mathcal{W}_{\text{diss}} = \int_{\Omega} -\frac{\partial D}{\partial \dot{d}} \delta \dot{d} \, dV, \quad (3)$$

with the dissipation function D . Here, we denote the body's volume by Ω , the body forces by $\mathbf{b}$, the traction vector by $\mathbf{t}$ acting on the Neumann boundary Γ_{σ} , and the rate of the damage parameter by $\dot{d}$. Using a damage function $f(d)$, the Helmholtz free energy Ψ is decomposed into a mechanical part and a regularization part, yielding

$$\Psi(\boldsymbol{\varepsilon}, f) := f \Psi_0(\boldsymbol{\varepsilon}) + \frac{1}{2} \beta |\nabla f|^2, \quad (4)$$

where β denotes the regularization parameter, and the mechanical energy of the undamaged material is given by

$$\Psi_0(\boldsymbol{\varepsilon}) = \frac{1}{2} \boldsymbol{\varepsilon} : \mathbb{E} : \boldsymbol{\varepsilon}, \quad (5)$$

with the elasticity tensor of order four $\mathbb{E}$ and the strains $\boldsymbol{\varepsilon} = \nabla^{\text{sym}} \mathbf{u} = \frac{1}{2}(\nabla \mathbf{u} + \mathbf{u} \nabla)$. For the damage function, several approaches can be chosen. The only requirements is that f decreases monotonically and $f \in]0, 1]$. Here, we chose $f = \exp(-d)$, such that $d \in [0, \infty[$ and $d = 0$ indicates the virgin material without any damage. Other damage models with the same approach can be found, for example, in Reference 28.

The only further part that needs to be defined is the dissipation function D . Here, the sole requirement is that $D = D(\dot{d}) \geq 0 \, \forall \dot{d}$. Possible choices are thus $D = r|\dot{d}|$ or $D_v = \eta \dot{d}^2/2$. Whereas D_v results in a rate-dependent, viscous evolution equation for the damage parameter d with the viscosity for damage η , D describes a rate-independent evolution. Since we are interested in brittle damage, we chose $D = r|\dot{d}|$ with a dissipation parameter r that is later identified as the energetic threshold for the onset of damage (evolution).

On the one hand, the evaluation of the stationarity conditions of Equation (1) provides the weak form of the balance of linear momentum, that is,

$$\int_{\Omega} \boldsymbol{\sigma} : \delta \boldsymbol{\varepsilon} \, dV - \int_{\Omega} \mathbf{b} \cdot \delta \mathbf{u} \, dV - \int_{\Gamma_{\sigma}} \mathbf{t} \cdot \delta \mathbf{u} \, dS = 0 \quad \forall \delta \mathbf{u}, \quad \mathbf{u} = \hat{\mathbf{u}} \quad \forall \mathbf{x} \in \Gamma_u, \quad (6)$$

where the mechanical stresses

$$\boldsymbol{\sigma} = \frac{\partial \Psi}{\partial \boldsymbol{\varepsilon}} = f \mathbb{E} : \boldsymbol{\varepsilon}, \quad (7)$$

have been introduced and displacements $\hat{\mathbf{u}}$ are prescribed on the Dirichlet boundary Γ_u . On the other hand, the stationarity condition for the damage parameter d in conjunction with Neumann condition $\mathbf{n} \cdot \nabla f = 0$ at the domain boundary results in

$$\dot{d} \geq 0, \quad \Phi := f \Psi_0 - \beta f \Delta f - r \leq 0, \quad \dot{d} \Phi = 0, \quad (8)$$

with the Laplace operator $\Delta f := \nabla \cdot \nabla f$. The first equation states the irreversibility of damage. The second equation shows that the quantity Φ serves as (local) indicator function for the onset and evolution of damage. It is thus comparable to a yield function well known from plasticity. More details on the model derivation are given in Reference 32, where a fast and stable algorithmic treatment for the equations has been presented. Unfortunately, the inequality (8) complicates the numerical evaluation to a large extent. Therefore, an operator split has been proposed in Reference 32, which simplifies the numerical treatment. The drawback is a distinct time-step dependency that can only be removed by choosing a fine temporal discretization. An adaptive choice of the time increments is thus unavoidable for a time-efficient procedure and will be employed in this work.

3 | NUMERICAL TREATMENT

3.1 | Discretization and operator split

For a numerical solution of Equations (6) and (8), a finite element procedure has been proposed in Reference 32. To be more precise, Equation (6) is discretized using the finite element approach

$$\mathbf{u} = \mathbf{S} \cdot \tilde{\mathbf{u}} \quad \text{and} \quad \delta \mathbf{u} = \mathbf{S} \cdot \delta \tilde{\mathbf{u}}, \quad (9)$$

where $\mathbf{S}$ denotes the shape functions, and $\tilde{\mathbf{u}}$ and $\delta \tilde{\mathbf{u}}$ denote the displacements and virtual displacements at the nodes, respectively. Furthermore, the $\mathbf{B}$ operator matrix is defined as usual by $\boldsymbol{\varepsilon} = \mathbf{B} \cdot \tilde{\mathbf{u}}$. Then, Equation (6) transforms to

$$\begin{aligned} \int_{\Omega} \delta \tilde{\mathbf{u}}^T \cdot \mathbf{B}^T \cdot \boldsymbol{\sigma} \, dV &= \int_{\Omega} \delta \tilde{\mathbf{u}}^T \cdot \mathbf{S}^T \cdot \mathbf{b} \, dV + \int_{\Gamma_{\sigma}} \delta \tilde{\mathbf{u}}^T \cdot \mathbf{S}^T \cdot \mathbf{t} \, dS \quad \forall \delta \tilde{\mathbf{u}} \\ \Rightarrow \int_{\Omega} \mathbf{B}^T \cdot \boldsymbol{\sigma} \, dV &= \int_{\Omega} \mathbf{S}^T \cdot \mathbf{b} \, dV + \int_{\Gamma_{\sigma}} \mathbf{S}^T \cdot \mathbf{t} \, dS. \end{aligned} \quad (10)$$

For the update of the damage field, the so-called neighbored element method has been used in Reference 32, which discretizes the Laplacian by means of selected neighbored elements. This procedure has been proven advantageous regarding computational costs. Unfortunately, we investigated some numerical sensitivity for distorted finite elements: the condition number of the algebraic system of equations increases dramatically, resulting in severe numerical errors. To avoid this drawback, we reviewed the neighbored element method and propose a different numerical treatment, taking all neighbors into account, as described in the following.

As for the original approach, we discretize the damage field with one value for each finite element, that is, we use a piecewise constant approximation on the finite element mesh Ω_h . Then, Equation (8) transforms to

$$f_e \bar{\Psi}_{0,e} - \beta f_e \frac{1}{|e|} \int_e \Delta f \, dV - r \leq 0 \quad \text{for all } e \in \Omega_h, \quad (11)$$

with the element's volume $|e|$ and the averaged elastic energy $\bar{\Psi}_{0,e} := \frac{1}{|e|} \int_e \Psi_0 \, dV$.

In order to approximate the Laplacian on an element e , we collect all nearest neighbors as shown in Figure 1. If a side of the element is located at the domain boundary, we take the Neumann zero conditions into account by mirroring the element at the boundary, and then enforcing the same value in the mirror element as for the element e itself by assigning the same element number to it. Finally, a set $\mathcal{N}_e$ of elements is constructed collecting all nearest neighbors, mirror elements and the element e itself. We now seek a stencil for the discrete Laplacian by Taylor expansion with the element barycenters $\mathbf{x}_e$ as

$$(\Delta_h f)_e = \sum_{e \in \mathcal{N}_e} a_{e\tilde{e}} f(\mathbf{x}_{\tilde{e}}) = \sum_{e \in \mathcal{N}_e} a_{e\tilde{e}} \left(f(\mathbf{x}_e) + \nabla f|_{\mathbf{x}_e} \cdot (\mathbf{x}_{\tilde{e}} - \mathbf{x}_e) + \frac{1}{2} (\mathbf{x}_{\tilde{e}} - \mathbf{x}_e)^T \cdot \mathbf{H}|_{\mathbf{x}_e} \cdot (\mathbf{x}_{\tilde{e}} - \mathbf{x}_e) + o(\|\mathbf{x}_{\tilde{e}} - \mathbf{x}_e\|^3) \right), \quad (12)$$

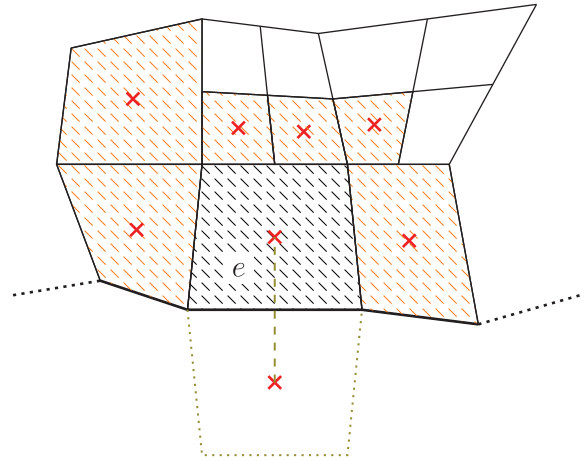


FIGURE 1 Neighbors (orange) and their barycenters (red cross) of an element e used to create the Laplacian approximation on an adaptively refined mesh. For a Neumann zero boundary condition, a mirror element at boundary edges is created (green)

with the Hessian matrix $\mathbf{H}$ of f . Rearranging the terms provides

$$\begin{aligned}
 (\Delta_h f)_e &= \left(\sum_{\tilde{e} \in \mathcal{N}_e} a_{e\tilde{e}} \right) f(\mathbf{x}_e) + \left(\sum_{\tilde{e} \in \mathcal{N}_e} a_{e\tilde{e}} (\mathbf{x}_e - \mathbf{x}_{\tilde{e}}) \right) \cdot \nabla f|_{\mathbf{x}_e} \\
 &\quad + \left(\sum_{\tilde{e} \in \mathcal{N}_e} a_{e\tilde{e}} \frac{1}{2} (\mathbf{x}_e - \mathbf{x}_{\tilde{e}})^2 \right) \partial_{xx}^2 f|_{\mathbf{x}_e} + \left(\sum_{\tilde{e} \in \mathcal{N}_e} a_{e\tilde{e}} \frac{1}{2} (\mathbf{y}_e - \mathbf{y}_{\tilde{e}})^2 \right) \partial_{yy}^2 f|_{\mathbf{x}_e} \\
 &\quad + \left(\sum_{\tilde{e} \in \mathcal{N}_e} a_{e\tilde{e}} (\mathbf{x}_e - \mathbf{x}_{\tilde{e}})(\mathbf{y}_e - \mathbf{y}_{\tilde{e}}) \right) \partial_{xy}^2 f|_{\mathbf{x}_e} \\
 &\quad + o(a_{e\tilde{e}}) o(\|\mathbf{x}_e - \mathbf{x}_{\tilde{e}}\|^3).
 \end{aligned} \tag{13}$$

The summation terms in brackets in front of the second derivatives must equal 1 while all other sums have to vanish. The term a_{ee} only appears in the first bracket and is thus determined once the other values are known. Locally numbering the N neighbors and mirror elements in $\mathcal{N}_e = \{e_0 := e, e_1, e_2, \dots, e_N\}$, we can thus solve a system of equations given by

$$\mathbf{D}^T \cdot \mathbf{a}_e = \mathbf{s}, \quad \text{with } \mathbf{D} \in \mathbb{R}^{N \times D}, \mathbf{a}_e \in \mathbb{R}^N, \mathbf{s} \in \mathbb{R}^D, \tag{14}$$

where D is the number of derivative terms in Equation (13) (ie, 5 in 2d) and the matrices and vectors are given by

$$\mathbf{D} := \begin{pmatrix} (x_1 - x_0) & (y_1 - y_0) & (x_1 - x_0)(y_1 - y_0) & \frac{1}{2}(x_1 - x_0)^2 & \frac{1}{2}(y_1 - y_0)^2 \\ (x_2 - x_0) & (y_2 - y_0) & (x_2 - x_0)(y_2 - y_0) & \frac{1}{2}(x_2 - x_0)^2 & \frac{1}{2}(y_2 - y_0)^2 \\ \vdots & & & & \vdots \\ (x_N - x_0) & (y_N - y_0) & (x_N - x_0)(y_N - y_0) & \frac{1}{2}(x_N - x_0)^2 & \frac{1}{2}(y_N - y_0)^2 \end{pmatrix}, \quad \mathbf{a}_e := \begin{pmatrix} a_{e,1} \\ a_{e,2} \\ a_{e,3} \\ \vdots \\ a_{e,N} \end{pmatrix}, \quad \mathbf{s} := \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}.$$

Assuming that $\mathbf{D}$ has full column rank, we can form the right-inverse of $\mathbf{D}^T$ given by $(\mathbf{D}^T)^{-1} = \mathbf{D} \cdot (\mathbf{D}^T \cdot \mathbf{D})^{-1}$ and the stencil entries are found by

$$\mathbf{a}_e = \mathbf{D} \cdot (\mathbf{D}^T \cdot \mathbf{D})^{-1} \cdot \mathbf{s}.$$

The system of equation in Equation (14) is underdetermined in most cases due to $N > D$, and thus one of many solutions is singled out. With the chosen stencil entries $\mathbf{a}_e$, the approximation of the Laplacian is finally given by

$$(\mathbf{A} \cdot \mathbf{f})_e := (\Delta_h f)_e = \partial_{xx}^2 f|_{\mathbf{x}_e} + \partial_{yy}^2 f|_{\mathbf{x}_e} + o(a_{e\tilde{e}}) o(\|\mathbf{x}_e - \mathbf{x}_{\tilde{e}}\|^3) = \Delta f|_{\mathbf{x}_e} + o(h), \tag{15}$$

provided that $o(a_{e\tilde{e}}) = o(h^{-2})$. Here, $\mathbf{f}$ denotes the vector of the discretized damage field, i.e., each entry of $\mathbf{f}$ corresponds to the damage value in the respective finite element. Please note that the Laplacian approximation depends solely on the mesh topology.

The method is identical to the one employed in Reference 32 for the case $N = D$, where the set of equations

$$\begin{aligned}
 f(\mathbf{x}_e) - f(\mathbf{x}_{\tilde{e}}) &= \nabla f|_{\mathbf{x}_e} \cdot (\mathbf{x}_e - \mathbf{x}_{\tilde{e}}) + \frac{1}{2} (\mathbf{x}_e - \mathbf{x}_{\tilde{e}})^T \cdot \mathbf{H}|_{\mathbf{x}_e} \cdot (\mathbf{x}_e - \mathbf{x}_{\tilde{e}}) \quad \text{for all } \tilde{e} \in S_e \\
 &\Leftrightarrow d\mathbf{f}_e = \mathbf{D} \cdot \partial \mathbf{f}_e,
 \end{aligned} \tag{16}$$

has been used to solve for the derivatives

$$\partial \mathbf{f}_e = (\mathbf{D}^T \cdot \mathbf{D})^{-1} \cdot \mathbf{D}^T \cdot d\mathbf{f}_e, \quad (17)$$

and then extracting the second order derivatives as

$$(\Delta_h f)_e = \partial \mathbf{f}_e \cdot \mathbf{s} = ((\mathbf{D}^T \cdot \mathbf{D})^{-1} \cdot \mathbf{D}^T \cdot d\mathbf{f}_e) \cdot \mathbf{s} = d\mathbf{f}_e \cdot \mathbf{D} \cdot (\mathbf{D}^T \cdot \mathbf{D})^{-1} \cdot \mathbf{s} = \mathbf{a}_e \cdot d\mathbf{f}_e. \quad (18)$$

Please note that an extension to the three-dimensional case is straight forward.

The advantage for the used Laplacian approximation is its direct applicability to adaptively refined meshes with hanging nodes taking into account all element neighbors. The alternative in Reference 32 requires a fixed number of neighboring elements (5 in 2d, 9 in 3d). Therefore, not all neighbors are taken into account even for a regular rectangular mesh, and this becomes even more important for adaptively-refined meshes with a couple of fine neighbors. Direct information flow within an iterative solution process would thus not be given for all direct neighbors. Discretization methods like the one employed here can be found, for example, in References 33 and 34.

With the above notation, we now have to devise a procedure to solve the inequality

$$\Phi_e := f_e \bar{\Psi}_{0,e} - \beta f_e (\mathbf{A} \cdot \mathbf{f})_e - r \leq 0, \quad \text{for all } e \in \Omega_h. \quad (19)$$

Intuitively, the solution behavior is as follows: for an undamaged domain, that is, $f \equiv 1$ and therefore Laplacian $(\mathbf{A} \cdot \mathbf{f})_e = 0$, the inequality will be automatically fulfilled for a small mechanical energy $f_e \bar{\Psi}_{0,e} < r$. Once the energy exceeds this threshold for an element e , the damage function f_e on the element has to be decreased to fulfill the inequality. However, due to a thereby-introduced difference to the element neighbors, now the Laplacian $(\mathbf{A} \cdot \mathbf{f})_{e_n} < 0$ is non-zero on the neighbor e_n , and this may hinder the fulfillment of the inequality for the neighbor. This is more severe the larger the regularization parameter β is chosen. Thus, the Laplacian value has to be mitigated by decreasing the damage function of neighbors as well. This way, the parameter β leads to a spreading of the damage or, equivalently, prevents steep gradients in the damage function.

3.1.1 | Elasticity solver

Given the current damage function $\mathbf{f}^{\hat{\mathbf{u}}}$, the displacement of the specimen for an increased loading given in terms of the prescribed displacements $\hat{\mathbf{u}} + d\hat{\mathbf{u}}$ has to be computed by solving Equation (10). To this end, we employ a geometric multigrid solver^{35,36} on a mesh hierarchy produced by successive refinement from an initial coarse finite element mesh. For an adaptive mesh hierarchy, we will use the adaptive multigrid approach^{37,38} without smoothing of the patch rims. In all simulations, we use a Gauss-Seidel smoother, V-cycle, three presmoothing and postsmoothing steps, the canonical prolongation and restriction, the RAP product for coarse matrices, and an LU base solver on the coarsest mesh. The multigrid is wrapped into a BiCGStab solver as preconditioner, and we solve for a residuum reduction or an absolute residuum of 10^{-10} (whatever applies first) with a maximal admissible iteration count of 200.

3.1.2 | Damage solver

Given the displacement of the specimen $\mathbf{u}^{\hat{\mathbf{u}}+d\hat{\mathbf{u}}}$ at the increased loading $\hat{\mathbf{u}} + d\hat{\mathbf{u}}$, the current damage distribution $\mathbf{f}^{\hat{\mathbf{u}}}$ has to be updated. To this end, we employ Algorithm 1 which is a Jacobi-like update scheme with an inner one-step Newton method: starting with the current damage values as initial guess, each element damage value is approximated by performing one Newton step for Equation (19) locally while keeping the required neighbor values of the Laplacian fixed. The local one-step Newton is used to solve a quadratic function in f_e . Since the step is only executed for a positive indicator function Φ_e , the damage value will strictly decrease and one step of the Newton will end up with an iterate above the root of the quadratic function. On the one hand, this shows the irreversibility of damage on the discrete level. On the other hand, this is quite beneficial for the overall iterative scheme, since neighboring values are to be updated one

after the other. The undershooting in the one-step Newton therefore allows converging quite smoothly to the solution. Note that the derivative of the Laplace operator solely depends on the mesh, and the matrix $\mathbf{A}$ can, therefore, be computed beforehand. Only on mesh adaption, the discrete Laplacian has to be updated. We accept the current iterate if the inequality (19) is fulfilled for each element up to a tolerance ϵ_{damage} and iterate up to a permitted maximal iteration count n_{damage} .

Algorithm 1. Damage update

Given: $\epsilon_{\text{damage}}, n_{\text{damage}}$ Input: $\mathbf{u}^{\hat{\mathbf{u}}+d\hat{\mathbf{u}}}, \mathbf{f}^{\hat{\mathbf{u}}}$	▷ numerical control parameters ▷ displacement at next load, current damage
On grid change: compute $\mathbf{A}$	▷ depends only on mesh topology
$\mathbf{f} := \mathbf{f}^{\hat{\mathbf{u}}}$	▷ set initial guess for damage
for $n = 1, 2, 3, \dots$, do Compute $\Lambda := \mathbf{A} \cdot \mathbf{f}$	▷ Laplacian of current iterate
for $e \in \Omega_h$ do $\Phi_e := f_e(\bar{\Psi}_{0,e} - \beta \Lambda_e) - r$	▷ value of the indicator function
if $\Phi_e \geq \epsilon_{\text{damage}}$ then $f_e := f_e - \frac{\Phi_e}{\bar{\Psi}_{0,e} - \beta(\Lambda_e + f_e(\mathbf{A})_{ee})}$	▷ if inequality violated ▷ one local Newton step
end if end for	
if $\ \Phi\ _{\max} \leq \epsilon_{\text{damage}}$ then break end if if $n > n_{\text{damage}}$ then abort end if	▷ accept iterate ▷ enforce maximal iteration count
end for	
$\mathbf{f}^{\hat{\mathbf{u}}+d\hat{\mathbf{u}}} := \mathbf{f}$	▷ accept last iterate

3.2 | Adaptivity

A crack-like evolution of a damage zone is highly localized both in space and in time, and it is thus advisable to address this behavior with adaptive numerical strategies for the computational mesh and for the load stepping.

3.2.1 | Loadstep control

For the loadstep control, we apply a simple heuristic: we decrease the load increment until the resulting change in the damage is reduced below a prescribed tolerance ϵ_{load} . As a measure for the change, we choose the maximum norm $\|\mathbf{f}^{\hat{\mathbf{u}}+d\hat{\mathbf{u}}} - \mathbf{f}^{\hat{\mathbf{u}}}\|_{\max} := \max_{e \in \Omega_h} |\mathbf{f}_e^{\hat{\mathbf{u}}+d\hat{\mathbf{u}}} - \mathbf{f}_e^{\hat{\mathbf{u}}}|$. This choice is preferable over other norms such as the Euclidian norm since it does not depend on the vector length, which will vary for mesh adaption and would lead to a weakened criterion. For the size of the loadstep increment, we impose maximal $d\hat{\mathbf{u}}_{\max}$ and minimal $d\hat{\mathbf{u}}_{\min}$ admissible bounds. Starting with a given

prediction $d\hat{\mathbf{u}}_{\text{predict}}$ (eg, $d\hat{\mathbf{u}}_{\text{predict}} = d\hat{\mathbf{u}}_{\text{max}}$ at the start of the simulation), we halve the size until the desired criterion is met. See Algorithm 2 for the implementation details.

Algorithm 2. Load step control

Given: $d\hat{\mathbf{u}}_{\text{min}}, d\hat{\mathbf{u}}_{\text{max}}, \epsilon_{\text{load}}$ ▷ numerical control parameters
 Input: $d\hat{\mathbf{u}}_{\text{predict}}, \mathbf{f}^{\hat{\mathbf{u}}}$ ▷ predicted step, computed damage at current load

$d\hat{\mathbf{u}} := d\hat{\mathbf{u}}_{\text{predict}}$ ▷ predicted stepsize as initial guess
loop
 Solve displacement $\mathbf{u}^{\hat{\mathbf{u}}+d\hat{\mathbf{u}}}$ for loading $\hat{\mathbf{u}} + d\hat{\mathbf{u}}$ given the current damage $\mathbf{f}^{\hat{\mathbf{u}}}$
 Solve damage $\mathbf{f}^{\hat{\mathbf{u}}+d\hat{\mathbf{u}}}$ for loading $\hat{\mathbf{u}} + d\hat{\mathbf{u}}$ given the computed displacement $\mathbf{u}^{\hat{\mathbf{u}}+d\hat{\mathbf{u}}}$
 if $\|\mathbf{f}^{\hat{\mathbf{u}}+d\hat{\mathbf{u}}} - \mathbf{f}^{\hat{\mathbf{u}}}\|_{\text{max}} < \epsilon_{\text{load}}$ **then break end if** ▷ accept solution
 if $d\hat{\mathbf{u}} = d\hat{\mathbf{u}}_{\text{min}}$ **then break end if** ▷ accept minimal stepsize
 $d\hat{\mathbf{u}} := \max(0.5 d\hat{\mathbf{u}}, d\hat{\mathbf{u}}_{\text{min}})$ ▷ reduce stepsize
end loop

$d\hat{\mathbf{u}}_{\text{predict}} := \begin{cases} d\hat{\mathbf{u}}, & \text{if stepsize reduced,} \\ \min(2 d\hat{\mathbf{u}}, d\hat{\mathbf{u}}_{\text{max}}), & \text{else} \end{cases}$ ▷ predict next stepsize

3.2.2 | Mesh size control

For the mesh size control, we employ adaptive mesh refinement before every load increment computation. Starting with an initial coarse mesh, we allow for the recursive refinement of mesh elements to increase the mesh resolution locally. The number of refinements that were applied to create a finite element is called the level of refinement of the element, and we allow refinements up to an a priori prescribed level $L_{\text{max}} \in \mathbb{N}$. For full refinement, all elements of the finite element mesh are refined. For adaptive refinement, however, only a selected subset of elements is chosen for further refinement, and some element patches may also be coarsened if admissible.

As an indicator for the required mesh resolution, we employ the elementwise value $\bar{\Psi}_e := f_e \bar{\Psi}_{0,e}$. Given that the elastic energy, $\bar{\Psi}_{0,e}$, is large, the element is likely to damage in the future and should be refined up to a proper resolution. If, however, the element is already damaged (small f_e) or the elastic energy is small, the mesh resolution can be kept small or even reduced for this element.

During the damage evolution, there are only a few localized elements where the indicator value $\bar{\Psi}_e$ is large compared to the bulk of elements with nonsignificant values. We therefore use the heuristic shown in Algorithm 3 for the selection of elements, which should be refined or coarsened: All elements with values below the scaled average $\theta_{\text{coarsen}}^{\text{avg}} \bar{\Psi}_{\text{avg}}$ are coarsened. Typical values of $\theta_{\text{coarsen}}^{\text{avg}}$ are 0.1 or below, thus removing elements only if their deviation from the average is really small. This is in particular the case for damaged elements once the damage process has finished locally. All elements with values above the scaled average $\theta_{\text{refine}}^{\text{avg}} \bar{\Psi}_{\text{avg}}$ are refined, using typical values for $\theta_{\text{refine}}^{\text{avg}}$ around 1.0. This leads to a detection of all elements that should be refined since they are more likely to damage than the average.

Upon mesh adaption, the relevant history data, that is, the damage, have to be transferred to the new mesh. For mesh refinement, the damage value of the parent element is used for all child elements. For mesh coarsening, the coarse element is assigned the averaged value of all the removed children since all children have the same element volume. For the displacements, we employ first-order Lagrangian shape functions. The arising hanging nodes due to mesh adaptivity are handled by interpolation constraints enforcing the continuity over element boundaries.

Algorithm 3. Mesh size control

Given: $\theta_{\text{refine}}^{\text{avg}}, \theta_{\text{coarsen}}^{\text{avg}}$ ($\theta_{\text{coarsen}}^{\text{avg}} < \theta_{\text{refine}}^{\text{avg}}$) ▷ numerical factors
Input: $\mathbf{f}, \Psi_0$ ($f_e, \Psi_0 \geq 0$) ▷ Damage function, elastic energy

$\bar{\Psi}_{\text{avg}} := \frac{1}{|\Omega_h|} \sum_{e \in \Omega_h} f_e \bar{\Psi}_{0,e} = \frac{1}{|\Omega_h|} \sum_{e \in \Omega_h} \bar{\Psi}_e$ ▷ current average of mechanical energy

for $e \in \Omega_h$ **do**
 if $\bar{\Psi}_e < \theta_{\text{coarsen}}^{\text{avg}} \bar{\Psi}_{\text{avg}}$ **then** ▷ coarsen elements below average
 coarsen(e)
 end if
 if $\bar{\Psi}_e > \theta_{\text{refine}}^{\text{avg}} \bar{\Psi}_{\text{avg}}$ **and** level(e) $< L_{\text{max}}$ **then** ▷ up to prescribed maximum
 refine(e) ▷ refine elements above average
 end if
end for

repeat ▷ additional refinements if required
 $\bar{\Psi}_{\text{avg}} := \frac{1}{|\Omega_h|} \sum_{e \in \Omega_h} f_e \bar{\Psi}_{0,e}$ ▷ update average of mechanical energy
 for $e \in \Omega_h$: **if** $\bar{\Psi}_e > \theta_{\text{refine}}^{\text{avg}} \bar{\Psi}_{\text{avg}}$ **and** level(e) $< L_{\text{max}}$ **then** refine(e) **end if**
until number of mesh vertices does not change

4 | NUMERICAL EXAMPLES

We illustrate the chosen method by two examples and study in detail the influence of several simulation parameters and the accuracy of the simulation that can be achieved. In particular, we are interested in the ability to resolve the sharp drop in a force-displacement plot for the tensile test by means of the adaptive strategies.

For the studies, we choose parameters as summarized in Table 1. While most parameters are fixed, we vary the regularization parameter β , the loading stepsize $d\hat{\mathbf{u}}$ (uniform) or its bounds $d\hat{\mathbf{u}}_{\min}, d\hat{\mathbf{u}}_{\max}$ (adaptive), and the maximal admissible level of grid refinement L_{max} for adaptive and fully refined meshing. Our simulations are carried out in 2d using the plain stress assumption.

Symbol	Quantity	Value	Unit
E	Young's modulus	200	GPa
ν	Poisson's ratio	0.33	—
r	Dissipation parameter	0.05	MPa
$\hat{\mathbf{u}}_{\text{end}}$	Final displacement on top edge	200	μm
β	Regularization parameter	$10^{-3}, \dots, 10^1$	MPa mm ²
$d\hat{\mathbf{u}}_{\min}$	Minimal load increment	$10^{-4}, \dots, 1$	μm
$d\hat{\mathbf{u}}_{\max}$	Maximal load increment	1	μm
ϵ_{load}	Loading step accuracy	10^{-2}	—
L_{max}	Max. level of refinement	0, ..., 5	—
$\theta_{\text{refine}}^{\text{avg}}$	Refinement factor	1.0	—
$\theta_{\text{coarsen}}^{\text{avg}}$	Coarsening factor	0.1	—
ϵ_{damage}	Damage iteration accuracy	10^{-8}	—
n_{damage}	Max. number of damage iterations	10^6	—

TABLE 1 Physical and simulation parameters

FIGURE 2 Single-notched geometry: A, Domain specification in mm and boundary conditions. B, Coarse finite element mesh. C, Full refinement for level 2. D, Adaptive refinement up to level 2 during damage propagation [Colour figure can be viewed at wileyonlinelibrary.com]

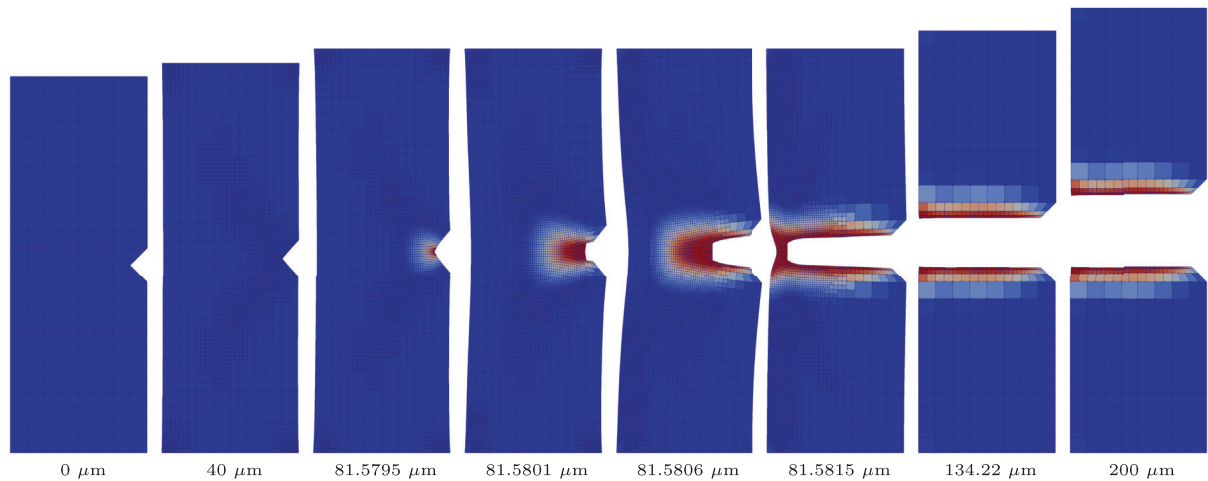
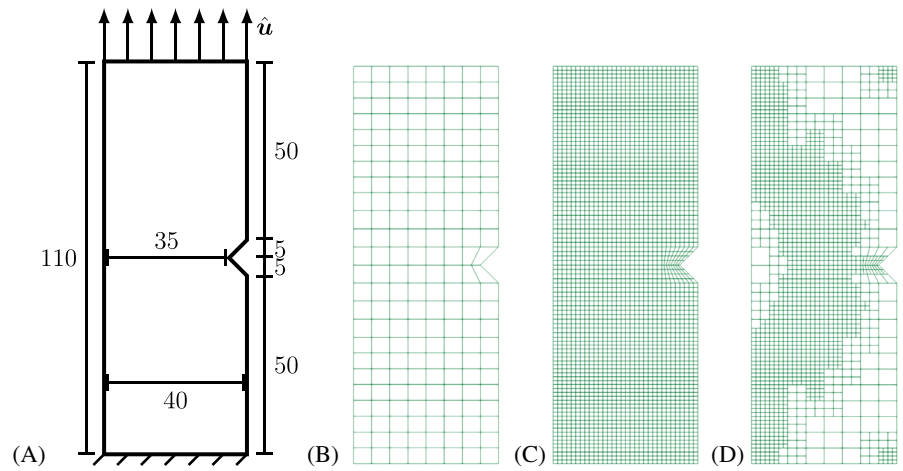


FIGURE 3 Single-notched geometry: displacement (100 times exaggerated) and damage function f (damaged parts in red) at different loading steps for a maximal level of refinement $L_{\max} = 3$, a minimal load increment $d\hat{u}_{\min} = 0.1$ nm, and $\beta = 1.0$ MPa mm². Damage values below 10^{-3} are considered as fully damaged and excluded from the visualization

4.1 | Single-notched geometry

The first example is a specimen with a single notch. The domain shape and size together with the chosen boundary conditions are depicted in Figure 2A. The domain is meshed by an initial coarse mesh (Figure 2B) and thereafter refined either using full (Figure 2C) or adaptive (Figure 2D) mesh refinement up to a prescribed level of refinement $L_{\max}$.

The result for a tensile test simulation up to a maximal displacement $\hat{u}_{\text{end}}$ of 200 μm at the top surface is shown in Figure 3 for selected loading positions employing a regularization parameter $\beta = 1.0$ MPa mm². As can be seen, the damage process starts at the notch, where largest stress values are to be expected, and then propagates through the specimen until it is split into two individually translating pieces. For this simulation, we used an adaptive strategy for both, load stepping and mesh refinement, with a minimal load increment $d\hat{u}_{\min} = 0.1$ nm and maximal mesh level $L_{\max} = 3$.

Measuring the force acting at the top of the specimen, we plot the computed force-displacement curve of this simulation run in Figure 4. The force drop is very sharply resolved and occurs at a displacement of roughly 73 μm with a maximal force of roughly 5.8 kN. The difference between adaptively and fully refined meshes can hardly be seen in the plot and a zoom is therefore added. Please note that the force drop difference between the two runs is only 0.1 μm , that is, a relative difference of 1.2%, which suggests that most of the spatial and load stepping resolution can be saved while maintaining the accuracy of the simulation within reasonable bounds. In the following, we will study this question in detail.

In order to illustrate the numerical behavior for the whole simulation range, we present the loading stepsize and the solver iteration counts in Figure 5. It can be nicely observed that the loadstep control starts with the maximal loading step $d\hat{u}_{\max}$, and large steps can be used until an initial damage formation is detected by the accuracy criterion. The stepsize is reduced down to $d\hat{u}_{\min}$ once the sharp force drop occurs. Thereafter, the stepsize can again be increased to its maximal

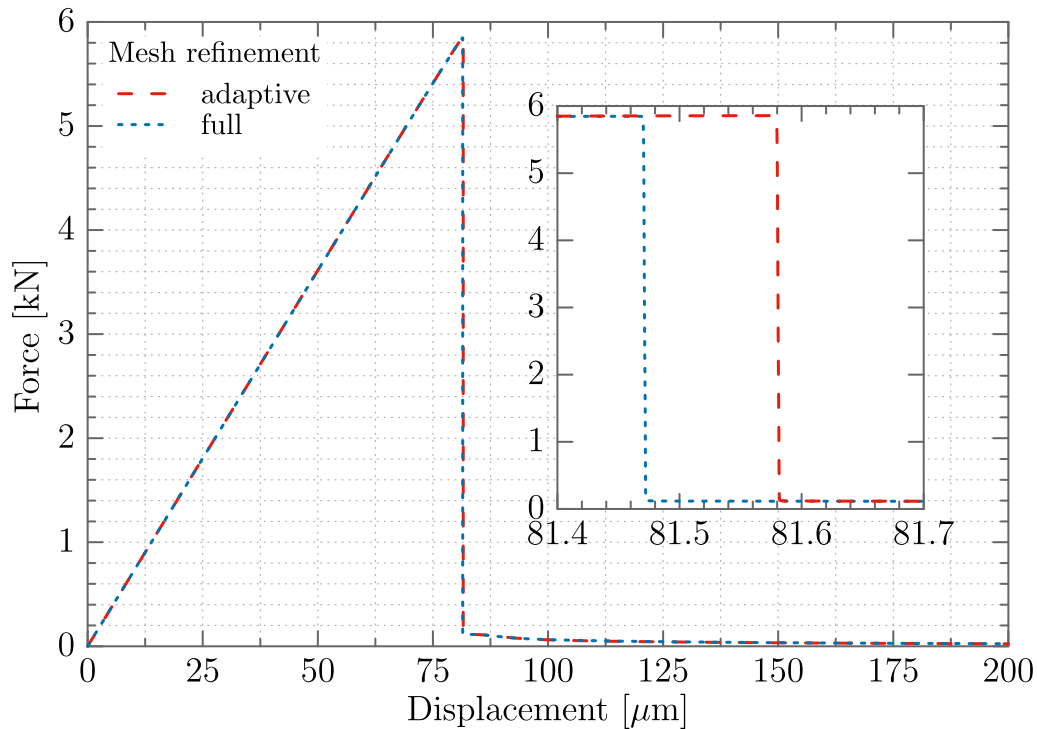


FIGURE 4 Single-notched geometry: force-displacement curve for a simulation with a maximal level of refinement $L_{\max} = 3$, a minimal load increment $d\hat{\mathbf{u}}_{\min} = 0.1\text{nm}$, and $\beta = 1.0\text{MPa mm}^2$ [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/nme.6549)]

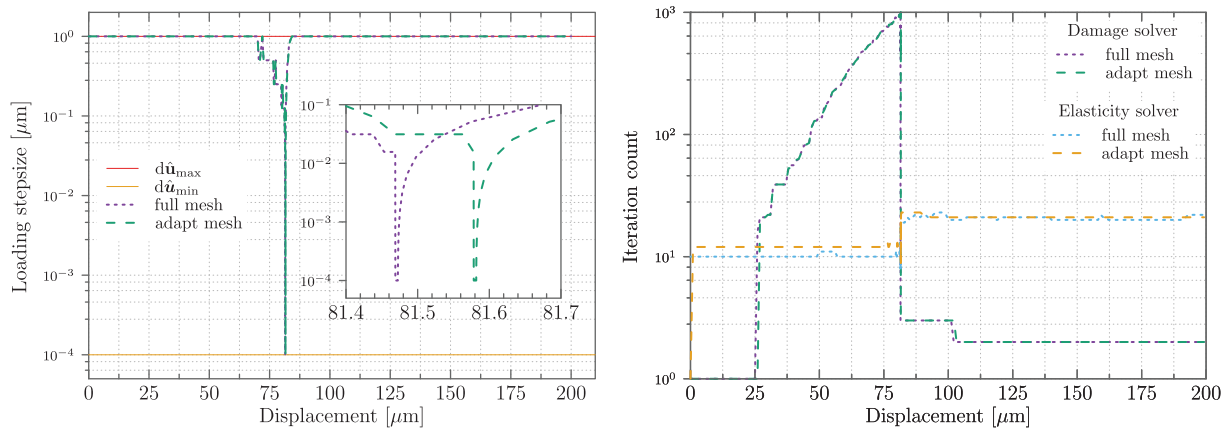


FIGURE 5 Single-notched geometry: loading stepsizes and iteration counts for damage update and elasticity solver for a simulation with a maximal level of refinement $L_{\max} = 3$, loading stepsize bounds $d\hat{\mathbf{u}}_{\min} = 0.1\text{nm}$ and $d\hat{\mathbf{u}}_{\max} = 1\text{μm}$, and $\beta = 1.0\text{MPa mm}^2$

admissible value. Taking a look at the solver iteration counts, we observe that the employed adaptive geometric multigrid solver shows small and bounded iterations counts over the whole range of the simulation. Note that the mesh and thereby the multigrid hierarchy are adapted with each loading step, and these numbers thus demonstrate a very robust behavior. The iteration count for the damage update is one when no damage formation occurs. Once the damage process initializes due to a sufficiently high mechanical energy, the iteration count remains in the order of 1000 and can actually drop due to smaller loading stepsizes.

In order to quantify the influence of the chosen numerical simulation parameters on the load at which the damage propagation and thereby the sharp force drop occurs, we present the results for adaptive and full mesh refinement at different values of ϵ_{load} and $d\hat{\mathbf{u}}_{\min}$ in Figure 6A. Further reducing the minimal loading stepsize $d\hat{\mathbf{u}}_{\min}$ by two orders of magnitude from 0.1 to 0.001 nm has only an influence of approximately 0.01μm, which is a 0.12‰ relative difference. Using a fully refined mesh instead of an adaptive refinement strategy has only an influence of approx. 0.1μm, that is, 1.4‰ relative difference. Increasing the required accuracy ϵ_{load} by an order of magnitude from 10^{-2} to 10^{-3} has only an influence of

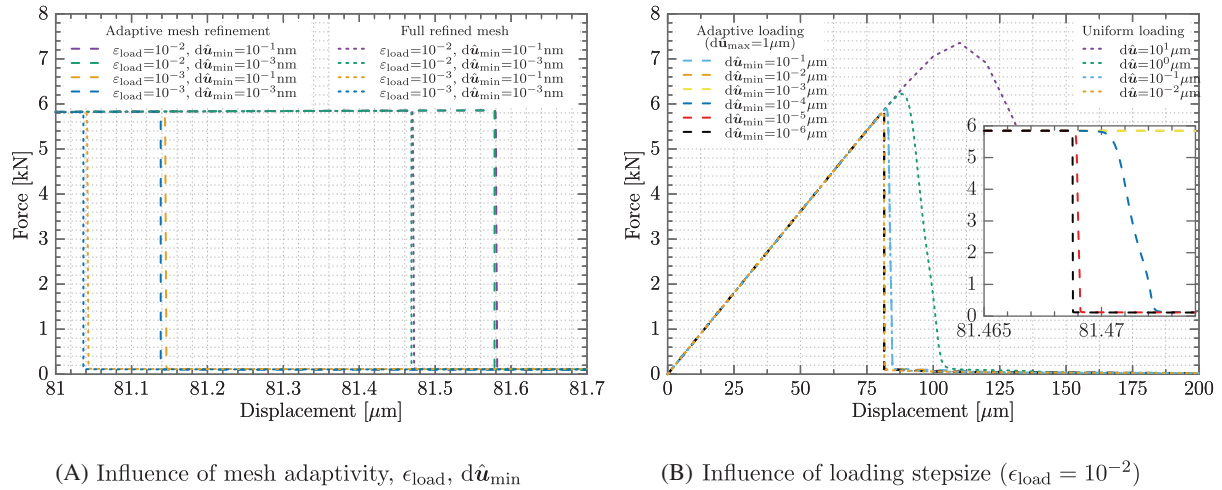


FIGURE 6 Single-notched geometry: influence study using $L_{\text{max}} = 3$ and $\beta = 1.0 \text{ MPa mm}^2$. A, Influence of mesh adaptivity ϵ_{load} , $d\hat{u}_{\text{min}}$. B, Influence of loading stepsize ($\epsilon_{\text{load}} = 10^{-2}$)

approx. $0.4\mu\text{m}$ which is 4.9% relative difference. We thereby conclude that the chosen parameters are sufficiently strict in order to produce reliable results, and we now turn our focus to the convergence properties with respect to mesh size, loading stepsize and the regularization parameter.

The influence of the loading stepsize is illustrated in Figure 6B. In this study, the uniform load stepping using the range $d\hat{u} = 10\mu\text{m}, \dots, 0.01\mu\text{m}$ is compared to an adaptive load stepping with a maximal load step $d\hat{u}_{\text{max}} = 1\mu\text{m}$ and minimal admissible load increments of size $d\hat{u}_{\text{min}} = 0.1\mu\text{m}, \dots, 10^{-6}\mu\text{m}$. For the coarse loading increments, the force-displacement curves differ significantly, and we thereby conclude that the stepsize has to be chosen sufficiently small. Comparing moderate loading stepsizes, there is not much difference between uniform and adaptive simulations of the same minimal stepsize. Therefore, adaptive stepping can be used in order to save execution time. Looking at the curves for very fine adaptive loading stepsizes, a convergence behavior is observed and there can hardly be seen a difference for the stepsizes below $d\hat{u}_{\text{min}} = 10^{-4}\mu\text{m} = 0.1\text{nm}$. In conclusion, we see that very fine load increments are required for the damage propagation phase in order to accurately capture the sharp force drop, but adaptive stepping is admissible for computational savings. We thus adopt the setup $d\hat{u}_{\text{min}} = 0.1\text{nm}$, $d\hat{u}_{\text{max}} = 1\mu\text{m}$ and adaptive load increments as a sufficient choice.

We now turn to the influence of mesh refinement. In Figure 7, the damage function is shown for selected loading steps, comparing the adaptively and fully refined meshing strategies. It can be nicely observed that the mesh adaptivity algorithm is capable of following the crack tip. More precisely, a fine resolution is induced already before the crack actually arrives at a certain position, and this allows for a high accuracy at the point of interest. We also see qualitatively the same picture for the mesh regions with the maximal refinement in both strategies. In the adaptive case, the coarsening of the mesh after damage formation leads to a coarser resolution after propagation. Of course, we could produce more similar pictures by simply not coarsening after damaging. However, our main objective is the force-displacement curve, and it will turn out that the coarsening does not influence the computed results.

In Figure 8, we compare the influence of using an adaptively refined mesh instead of full refinement. Curves of the same color correspond to the same maximal spatial refinement level, that is, in the highly resolved regions, the element size is equal, while the element size is significantly larger in other parts of the domain for adaptive meshing. It can be observed that full refinement and adaptive mesh handling produce very similar curves for all refinement levels. Typically, the curves of full refinement are slightly more to the left due to the slightly better accuracy. For the adaptive refinement, a convergence behavior is observed for the curves. We therefore conclude that for this example and using $\beta = 1 \text{ MPa mm}^2$, the choice $L_{\text{max}} = 3$ or below is sufficient for a high accuracy of the simulation. In particular, adaptive meshing is admissible.

The employed strategy in Algorithm 2 implements an operator split. The equation for the displacements and the equation for the damage function are solved one after the other in every loadstep. Once the change in the damage function is small enough, we simply proceed to the next loadstep without rechecking that both parts, displacements and damage, have converged simultaneously. In order to quantify the amount of additional error introduced by this operator splitting, we also compute the force-displacement curves with an additional outer loop that checks for a simultaneous

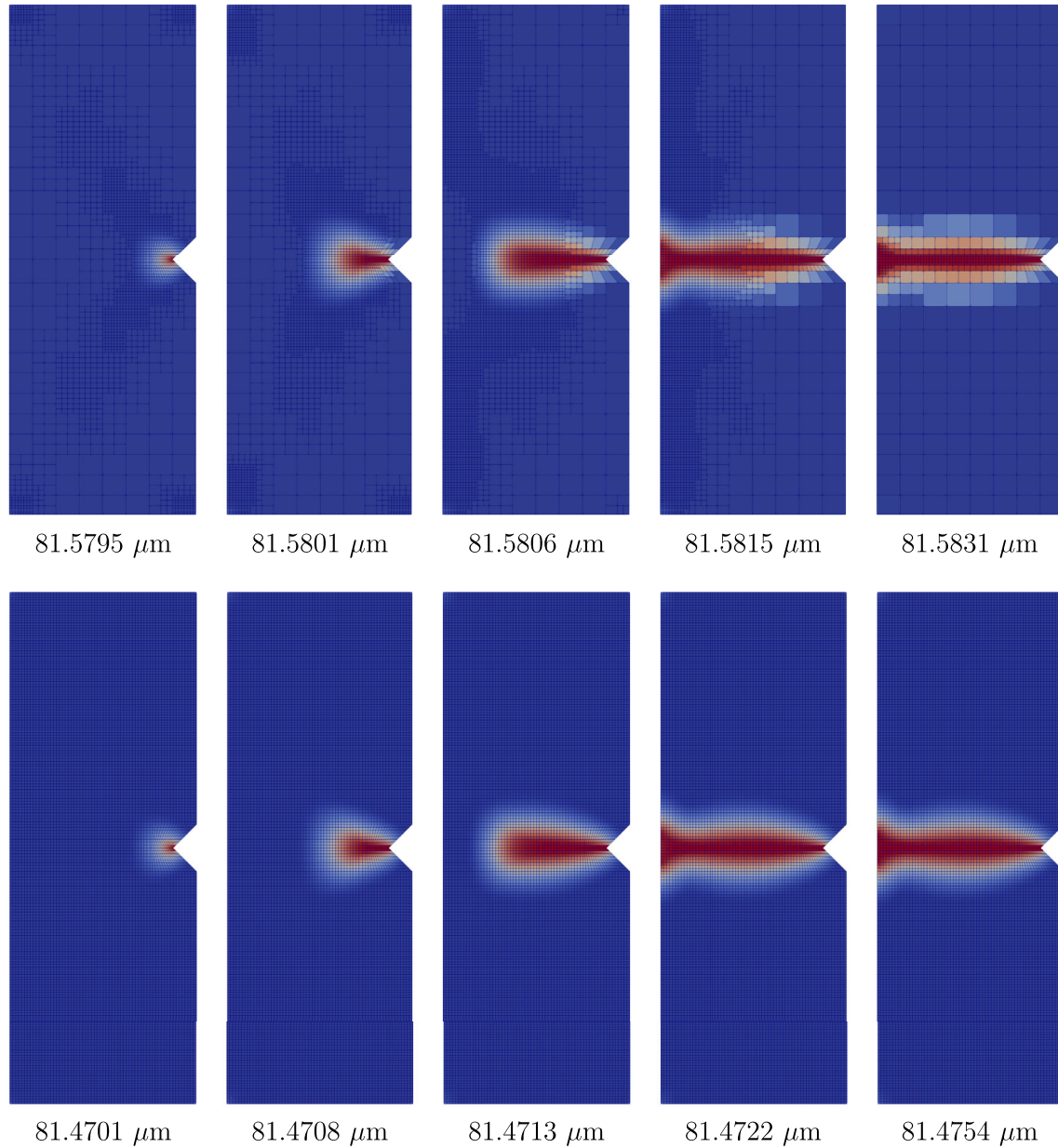


FIGURE 7 Single-notched geometry: adaptive (top) and fully (bottom) refined solution for the damage function f for $L_{\max} = 3$, $d\hat{u}_{\min} = 0.1 \text{ nm}$, $d\hat{u}_{\max} = 1 \text{ } \mu\text{m}$, $\beta = 1.0 \text{ MPa mm}^2$ for various load steps [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/nme.6549)]

reduction of the error in both equations. To this end, we iteratively compute the solution for displacement $\mathbf{u}^{\hat{\mathbf{u}}+d\hat{\mathbf{u}}}$ and damage $\mathbf{f}^{\hat{\mathbf{u}}+d\hat{\mathbf{u}}}$ while fixing the other solution until they both differ from the previous iterate as $\|\mathbf{f}_k^{\hat{\mathbf{u}}+d\hat{\mathbf{u}}} - \mathbf{f}_{k-1}^{\hat{\mathbf{u}}+d\hat{\mathbf{u}}}\|_{\max} + \|\mathbf{u}_k^{\hat{\mathbf{u}}+d\hat{\mathbf{u}}} - \mathbf{u}_{k-1}^{\hat{\mathbf{u}}+d\hat{\mathbf{u}}}\|_{\max} < \epsilon_{\text{cpl}}$ ($k \geq 2$). This requires at least two solutions of the system. Once the coupled system has been solved to this accuracy, we still require the change in damage to be below the threshold $\|\mathbf{f}^{\hat{\mathbf{u}}+d\hat{\mathbf{u}}} - \mathbf{f}^{\hat{\mathbf{u}}}\|_{\max} < \epsilon_{\text{load}}$ for the control of the loadstep size. For the mesh adaptive case, we also adapt the mesh within the iteration loop for the iterative coupling. The results of the study are presented in Figure 9. The force-displacement curves are very close in all cases. A zoom has been added to Figure 9A to visualize the difference that is at most $0.5 \text{ } \mu\text{m}$ or 0.6% . The plotted marks indicate one loadstep computation for the different runs. While the force drop is resolved by a lot of load steps in the case of the operator split, the drop is resolved by one loadstep in the iterative coupling strategy. To this end, the iteration count within one loadstep increases drastically at the force drop as can be observed in Figure 9B. The loading stepsize is decreased similarly in most cases as shown in Figure 9C, except for a tightened loading accuracy that leads to an earlier decrease of the stepsize. In general, the operator split only introduces a small error. Compared to the workload of the iterative coupling, it is either iterating within a loadstep or iterating over several loadsteps to resolve the force drop.

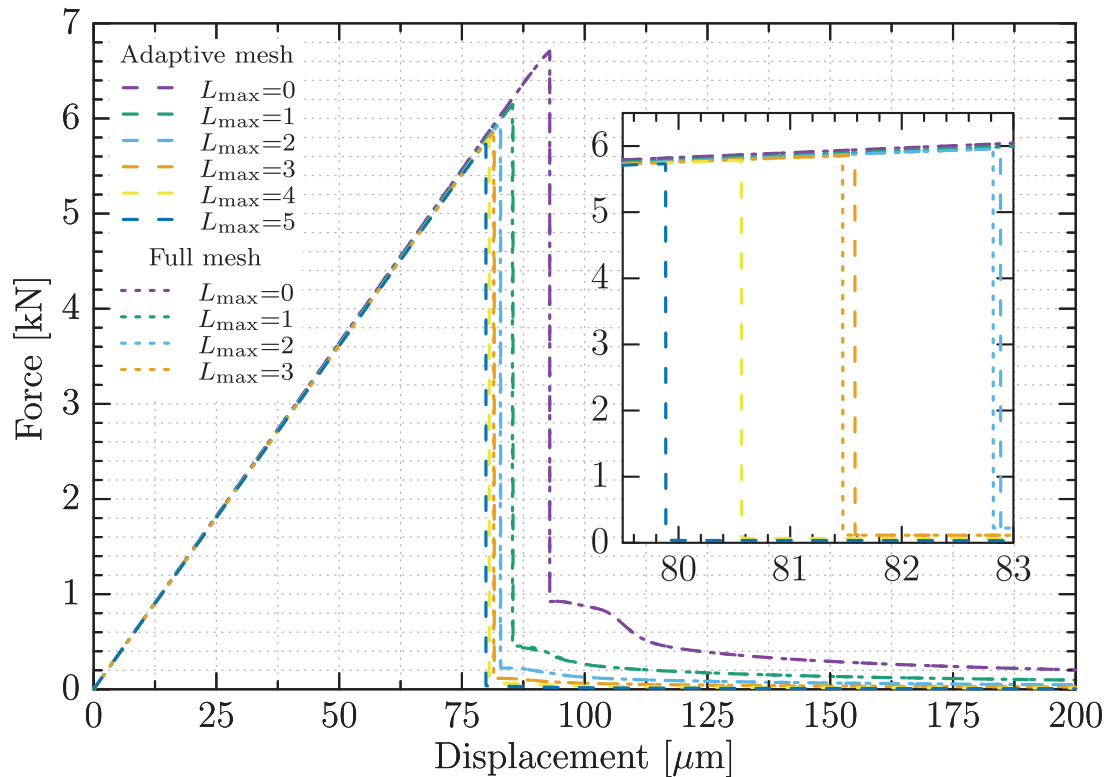


FIGURE 8 Single-notched geometry: comparison of full and adaptive mesh refinements for a simulation with $d\hat{u}_{\min} = 0.1\text{ nm}$, $d\hat{u}_{\max} = 1\text{ }\mu\text{m}$, $\beta = 1.0\text{ MPa mm}^2$

We are now in the position to own a reliable, highly accurate simulation tool for the force-displacement curve of the chosen brittle damage model. Using sufficiently fine load stepping and mesh refinement, we can accurately compute the position of the force drop in a tensile test. The sensitivity with respect to the numerical parameter has been confirmed to be below reasonable bounds. The remaining unstudied parameters are the dissipation parameter r , which we keep fixed in this contribution, and the regularization parameter β of the brittle material model. In Figure 10A, we have computed the force-displacement curves for different choices of β and different mesh refinement levels. For large regularization values, $\beta = 10, 1\text{ MPa mm}^2$, a convergence of the curves is observed. For $\beta = 0.1\text{ MPa mm}^2$, level 3 is still not sufficient and convergence is observed starting at level 4. For $\beta = 0.01, 0.001\text{ MPa mm}^2$, a convergence of the curve is not observed. Taking a look at the corresponding damage functions at crack initialization depicted in Figure 10B, this is in agreement with the observation that the crack width has to be resolved by more than one element. With decreasing regularization parameter β , the gradient of the damage function is less penalized and therefore the crack width decreases. Thus, more refined meshes have to be used for smaller regularization parameters. For the non-converging cases, the crack is resolved only by a single line of elements which is obviously too coarse.

Figure 11 presents a study for the influence of the specimen size. The size of the notch is kept constant, but three different meshes (Figure 11A) with different length are computed. An equal displacement on one side is enforced for all three simulations. The resulting strain is therefore smaller for larger specimen size. Consequently, the damage occurs at a later displacement as can be observed in Figure 11B.

Finally, we take a look at the runtime behavior for our simulations in Figure 12. All simulations have been performed on a laptop with a 2.9 GHz Intel Core i7-7820HQ processor, 16GB main memory and single core execution. The implementation has been realized within the finite element simulation framework ug4.³⁹ Coarse grid meshing employs ProMesh,⁴⁰ visualization is by Paraview.⁴¹

Since the simulation allows for loading steps in different orders of magnitude, using adaptive load stepping clearly offers a large potential for computational savings. This is shown in Figure 12A, where the runtimes for uniform load step-sizes quickly grow to the order of a couple of hours already at a coarse loading resolution of 10 nm. An even smaller stepsize is thus prevented for uniform simulations. However, using adaptive resolution of the load increments, the simulations can be carried out with runtimes below 100 seconds to achieve a very high accuracy at the desired spots while taking larger steps where permitted.

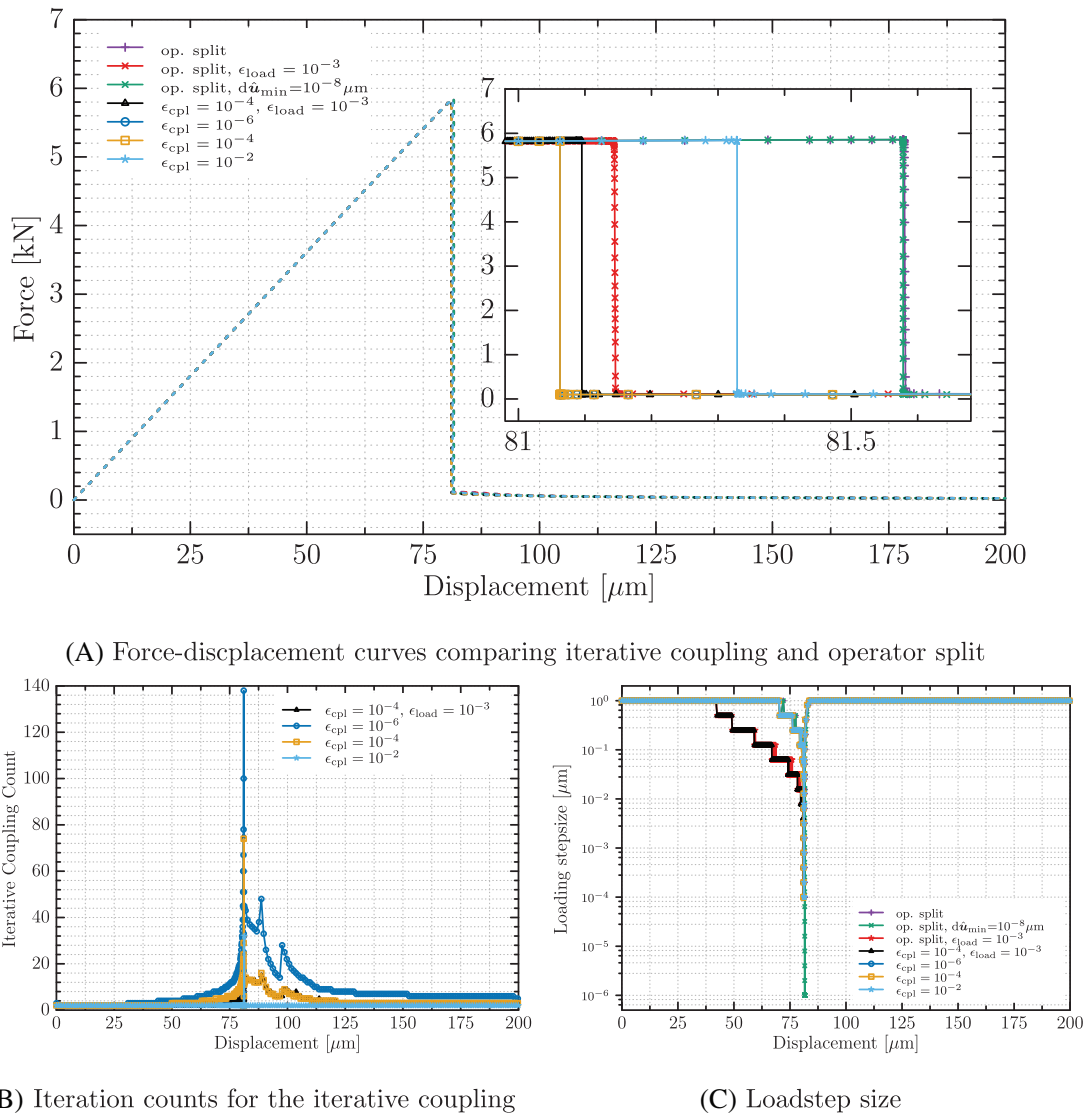
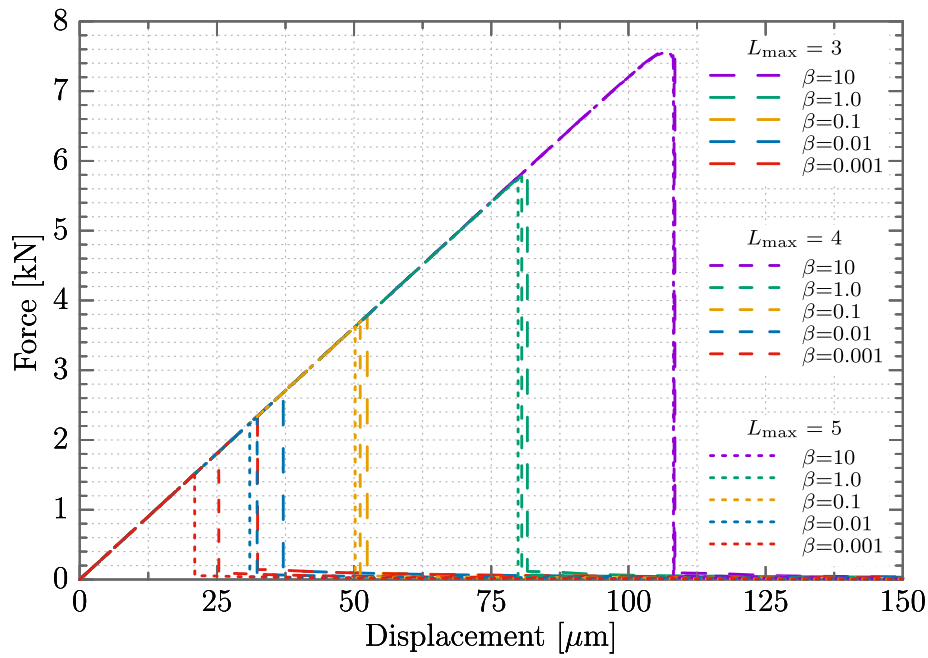
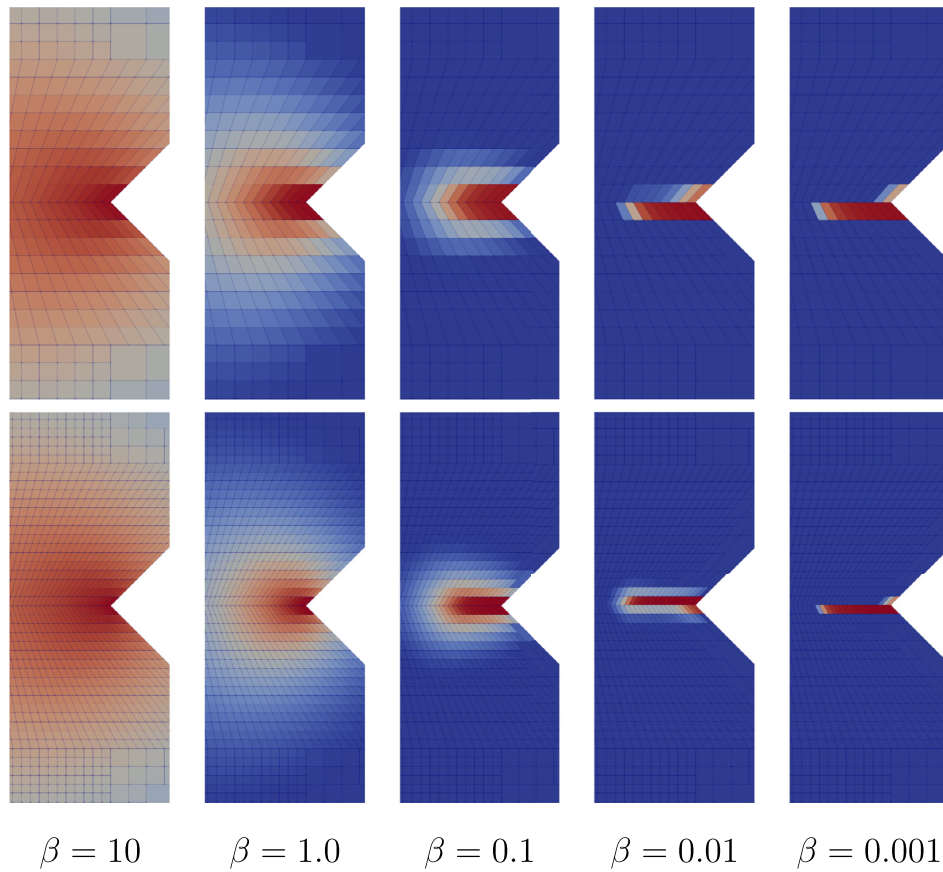


FIGURE 9 Single-notched geometry: Influence study for the iterative coupling using $L_{\text{max}} = 3$, $\beta = 1.0 \text{ MPa mm}^2$, $d\hat{u}_{\text{max}} = 1 \mu\text{m}$, $d\hat{u}_{\text{min}} = 0.1 \text{ nm}$, and $\epsilon_{\text{load}} = 10^{-2}$ if not otherwise stated. A, Force-displacement curves comparing iterative coupling and operator split. B, Iteration counts for the iterative coupling. C, Loadstep size

One benefit of using adaptive mesh refinement is a substantial saving in memory requirement. This allows us to run simulations with a locally high resolution for adaptive simulations, whereas fully refined meshes with the same resolution are prevented due to memory constraints. A second potential benefit for adaptive simulations is runtime savings. However, the gain due to smaller element numbers has to be compared to the mesh adaption overhead. In Figure 12B (left), we show the runtime for several components of our simulation. Presented is the overall simulation time for adaptively and fully refined computations. For the adaptive meshing, we also list the accumulated fraction of time spent for mesh adaption and for the amount of the simulation time consumed for the computation of the loadsteps. Each step further subdivides into the time spend for the elasticity solver and the damage solver. Since the damage solver requires a reinitialization of the discrete Laplacian, we also list the timings for the matrix setup. All times are accumulated numbers for the individual parts over a whole simulation run. For the fully refined case, we report overall simulation time and the fractions spend for solving elasticity and damage, while the Laplacian setup cost is neglectable due to the missing mesh adaption. It can be observed that for small mesh resolutions with runtimes in the order of seconds, the full refinement is comparable in runtime due to the grid adaption overhead. Also note that the damage solver requires a new neighbor search only for the adaptive case. Nevertheless, the picture changes at around refinement level 3, where the adaptive run is significantly faster than full refinement. At refinement level 5, an adaptive simulation is still in the order of minutes while the full

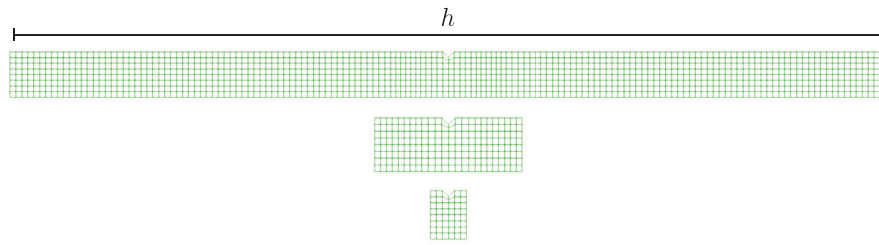


(A) Force-displacement curves under mesh refinement for different values of β

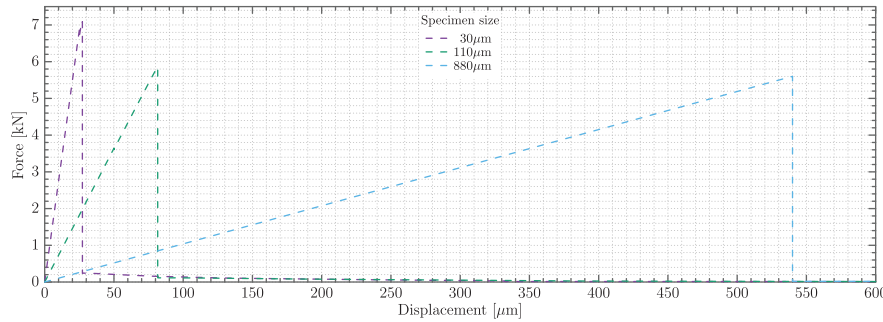


(B) Damage function f (zoom at notch) for $L_{\max} = 3$ (top) and 4 (bottom)

FIGURE 10 Single-notched geometry: comparison of different regularization parameter β (in MPa mm^2) using adaptive mesh refinement, $d\hat{\mathbf{u}}_{\min} = 0.1\text{nm}$, $d\hat{\mathbf{u}}_{\max} = 1\text{ }\mu\text{m}$. A, Force-displacement curves under mesh refinement for different values of β . B, Damage function f (zoom at notch) for $L_{\max} = 3$ (top) and 4 (bottom) [Colour figure can be viewed at wileyonlinelibrary.com]



(A) Specimens with different length: $h = 880\mu\text{m}$ (top), $h = 110\mu\text{m}$ (middle), $h = 30\mu\text{m}$ (bottom)



(B) Force-displacement for regularization parameter $\beta = 1 \text{ MPa mm}^2$, adaptive mesh refinement, $d\hat{u}_{\min} = 0.1\text{nm}$, $d\hat{u}_{\max} = 1\mu\text{m}$

refined simulation takes several hours. The increase in runtime can be mainly attributed to a growth in iteration numbers for the damage update as reported in Figure 12B (right). The employed multigrid solver for the elasticity problem remains quasi-constant under mesh refinement. For both solvers, we show the average of the required number of iterations over a whole simulation run as well as the maximum.

The runtime behavior with respect to the regularization parameter β is shown in Figure 12C (left). Both, adaptive and full mesh refinement show similar runtime behavior for the whole range of the regularization parameters except for the case $\beta = 10 \text{ MPa mm}^2$, where the damage is widely smeared out and runtimes explode for refinement level 5. In Figure 12C (right), we report the average iteration count for the damage solver under mesh refinement. In all cases, the iteration count increases substantially with mesh refinement due to the increased number of unknowns and the chosen Jacobi-like solver. However, due to the fast execution of a single iterate, the runtime for the damage update is comparable to elasticity solver even on mesh level 5.

4.2 | Double-notched geometry

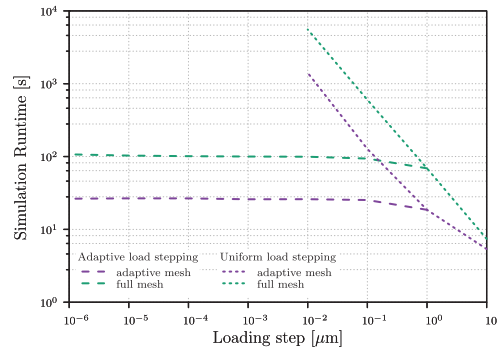
As a second example, we employ a double-notched specimen as depicted in Figure 13A and again the simulation parameters as shown in Table 1. A similar test has already been presented in Reference 32. The coarse grid meshing (Figure 13B) is already performed with a slightly higher resolution at the notches.

One particularity arises for the curved boundary at the notches, which is an ideal half-circle. With finer full (Figure 13C) or adaptive (Figure 13D) meshing, the circle can be better resolved, and we therefore project newly-created mesh nodes onto the boundary description. In order to prevent very distorted meshes by this projection, we also move the nodes in a neighborhood of the notch boundary during mesh refinement. Also note that, using the boundary projection, a slightly different problem is solved numerically for every mesh refinement level since a finer and finer polygonal approximation is used. For this reason, we keep the mesh refinement at the highest level close to the notch boundary for the whole simulation run for adaptive simulations as well.

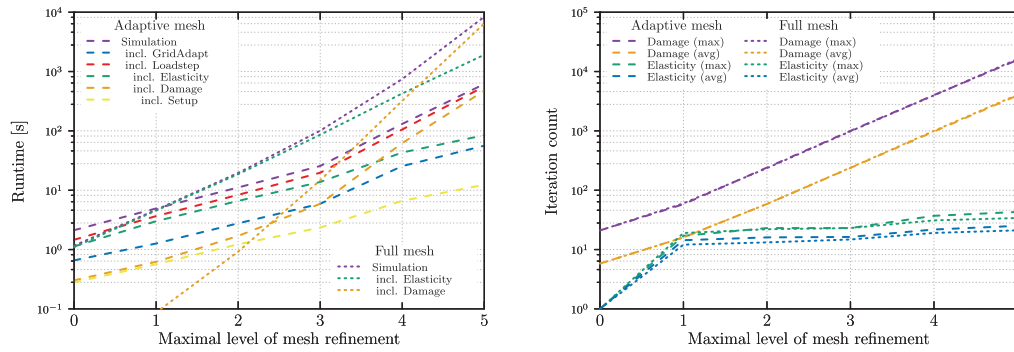
In Figure 14, we show the simulated results of a tensile test up to $200\mu\text{m}$ displacement. The damage evolution starts at the notches and propagates through the specimen until only very few finite elements join the three separated parts of the original body. Since the few elements show almost no stiffness against bending, this state has thus some similarities known for plastic joints. For this example, we have chosen a slightly smaller refinement factor $\theta_{\text{refine}}^{\text{avg}} = 0.8$ such that already some more elements even below the average are refined while we keep the coarsening factor $\theta_{\text{coarsen}}^{\text{avg}} = 0.1$ as before. In Figure 15, we compare the damage propagation for full and adaptive meshing strategies and see that they are in a good qualitative agreement.

FIGURE 11

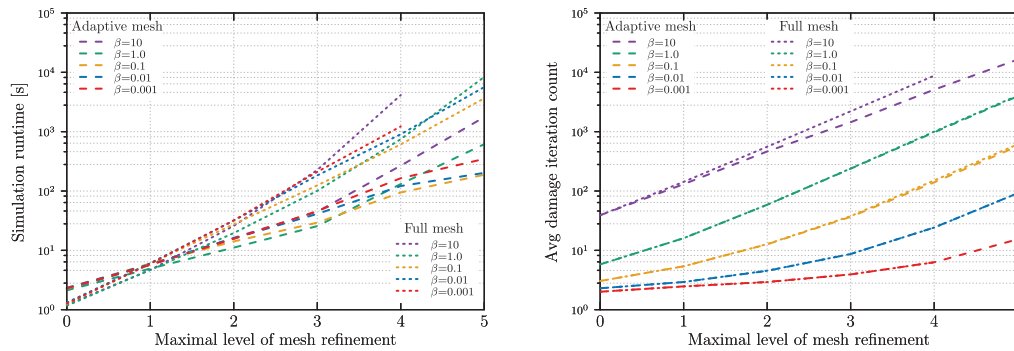
Single-notched geometry: comparison of specimens with different length. A, Specimens with different lengths: $h = 880 \mu\text{m}$ (top), $h = 110 \mu\text{m}$ (middle), $h = 30 \mu\text{m}$ (bottom). B, Force-displacement for regularization parameter $\beta = 1.0 \text{ MPa mm}^2$, adaptive mesh refinement, $d\hat{u}_{\min} = 0.1\text{nm}$, $d\hat{u}_{\max} = 1 \mu\text{m}$



(A) Comparison of adaptive and uniform loading stepsizes for $L_{\max} = 3$ and $\beta = 1.0 \text{ MPa mm}^2$



(B) Runtimes and iteration counts under mesh refinement for $\beta = 1.0 \text{ MPa mm}^2$ and $d\hat{u}_{\min} = 0.1\text{nm}$



(C) Simulation runtimes and iteration counts for varying β using $L_{\max} = 3$ and $d\hat{u}_{\min} = 0.1\text{nm}$

FIGURE 12 Single-notched geometry: runtimes (excluding output of solution to hard disc and statistic computations) and iteration counts for the damage and elasticity solver. A, Comparison of adaptive and uniform loading stepsizes for $L_{\max} = 3$ and $\beta = 1.0 \text{ MPa mm}^2$. B, Runtimes and iteration counts under mesh refinement for $\beta = 1.0 \text{ MPa mm}^2$ and $d\hat{u}_{\min} = 0.1\text{nm}$. C, Simulation runtimes and iteration counts for varying β using $L_{\max} = 3$ and $d\hat{u}_{\min} = 0.1\text{nm}$

The comparison of the force-displacement curves for full and adaptive refinement is shown in Figure 16. We can again observe a similar force drop between the two runs. Figure 17 shows the adaptive loading stepsizes and the solver iteration counts. For this example, the damage propagation is not in a single process but happens in certain steps. The loading increment has thus to be reduced adaptively a couple of times, but can be increased to large values in between. The iteration counts for the damage solver as well as the elasticity solver are bounded for the whole simulation range with smaller iteration numbers at small loading stepsizes.

Looking at the convergence for mesh refinement in Figure 18A, we find all curves to be close to each other. The comparison for different regularization parameters in Figure 18B again shows the large influence of β for the crack

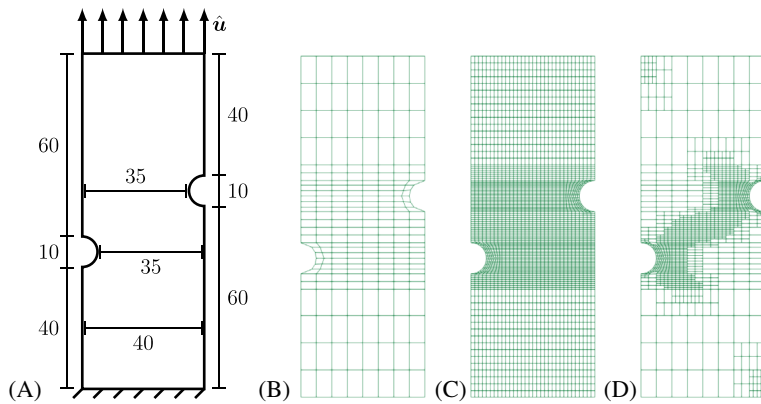


FIGURE 13 Double-notched geometry: A, Domain specification in mm and boundary conditions. B, Coarse finite element mesh. C, Full refinement for level 2. D, Adaptive refinement up to level 2 during damage propagation [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/nme.6549)]

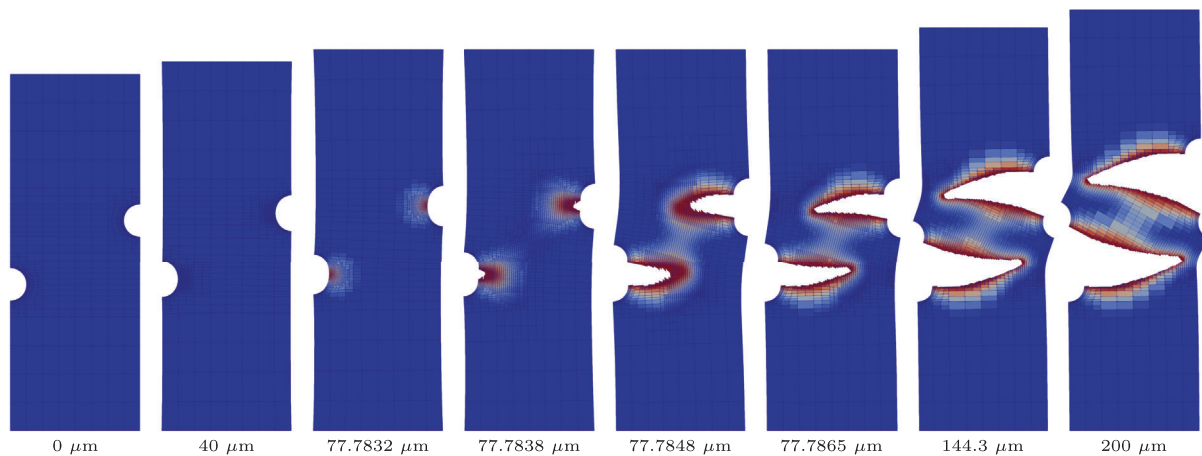


FIGURE 14 Double-notched geometry: displacement (100 times exaggerated) and damage function f (damaged parts in red) at different loading steps for a maximal level of refinement $L_{\max} = 3$, a minimal load increment $d\hat{u}_{\min} = 0.1 \text{ nm}$, and $\beta = 1.0 \text{ MPa mm}^2$. Damage values below 10^{-3} are considered as fully damaged and excluded from the visualization

loading. For all β values, a convergence can be observed in this example, which is in agreement with a visual inspection in Figure 18C.

In Figure 19, we investigate the sensitivity of our numerical method with respect to symmetries. However, a symmetric damage propagation is observed for an entirely symmetric problem description and meshing, changing a single node of the boundary description or in the interior of the mesh results in a nonsymmetric damage evolution. This is in accordance to the observation that the symmetric solution is an instable path and the nonsymmetric solution will be triggered in a real-world specimen due to imperfections and local fluctuations.

In Figure 20, we present the behavior of the model employing arc length control. For this study, an implementation of the model in the finite element code FEAP⁴² has been used without adaptive load stepping and mesh adaptation. The curve length is resolved using two different step counts such that the arc length increment differs by a factor of two. The results show that there exist no remarkable differences in the curves and the structural toughness for the different arc length increments. Furthermore, they reveal that an asymmetric distribution for the damage function is obtained even for a symmetric finite element mesh. Although damage initialization is detected at both notches, only at the upper notch, a distinct crack evolves. This behavior is accompanied by a snapback in the force-displacement diagram (Figure 20). After reaching this maximum value of the reaction force, a rapid reduction both in force and displacement can be observed which is almost parallel to the ascending loading. The force-displacement diagram shows that the dissipated energy, which equals the area of the hysteresis, is much smaller than the apparent dissipated energy in our previous results, in which the dissipated energy equals approximately the area of a triangle. This is due to the displacement-controlled boundary conditions. This suggests employing arc length control with the presented adaptive meshing and loading strategies in future works.

FIGURE 15 Double-notched geometry: adaptive (top) and fully (bottom) refined solution for the damage function f for $L_{\max} = 3$, $d\hat{\mathbf{u}}_{\min} = 0.1\text{nm}$, $d\hat{\mathbf{u}}_{\max} = 1\text{ }\mu\text{m}$, $\beta = 1.0\text{ MPa mm}^2$ for various load steps

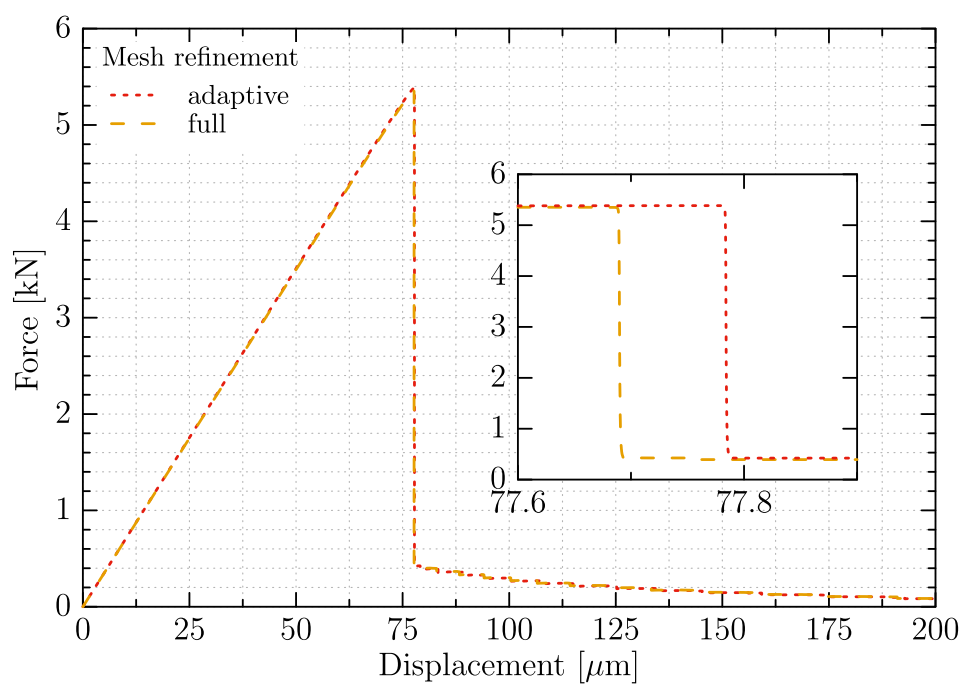
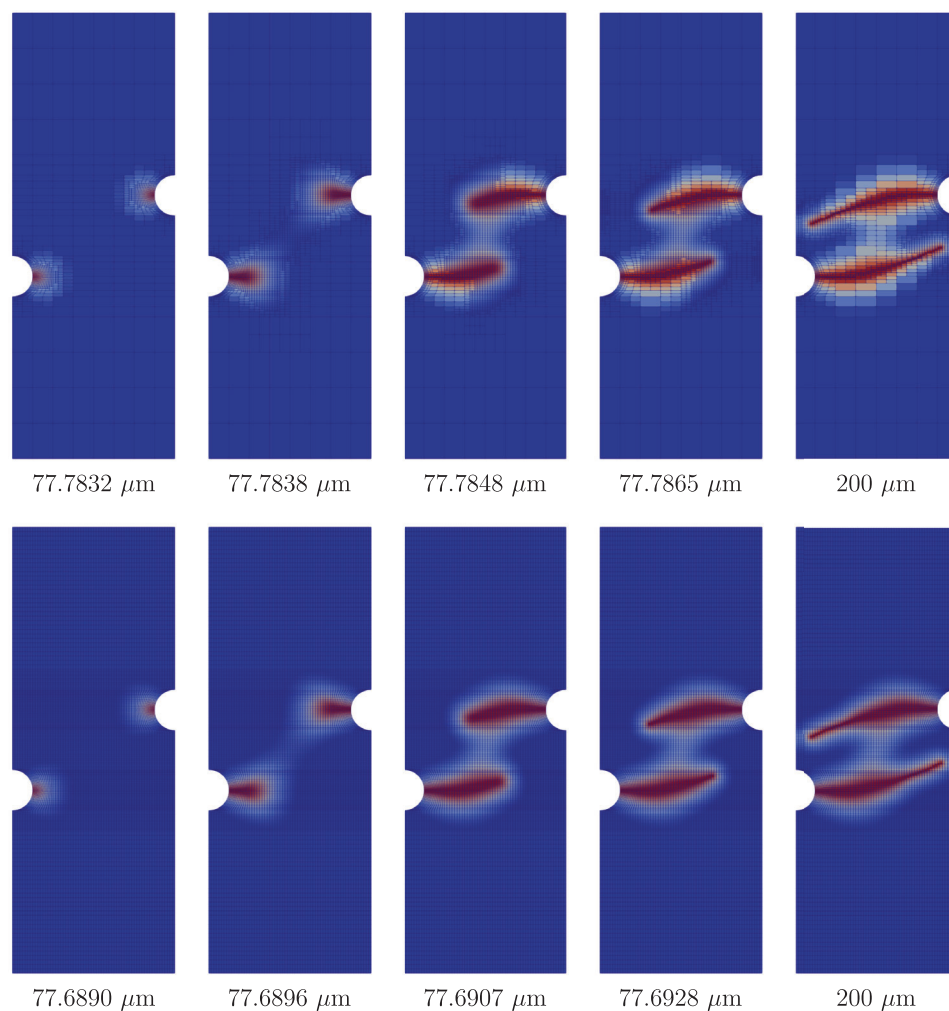


FIGURE 16 Double-notched geometry: force-displacement curve for a simulation with a maximal level of refinement $L_{\max} = 3$, a minimal load increment $d\hat{\mathbf{u}}_{\min} = 0.1\text{nm}$, and $\beta = 1.0\text{ MPa mm}^2$ [Colour figure can be viewed at wileyonlinelibrary.com]

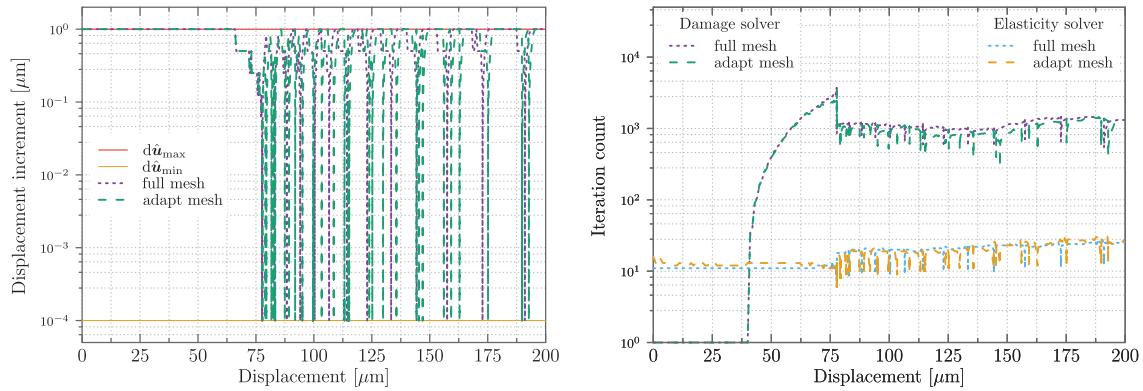
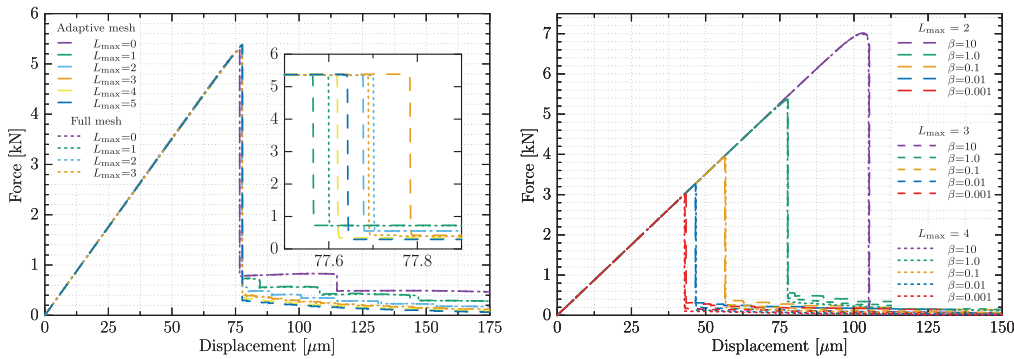
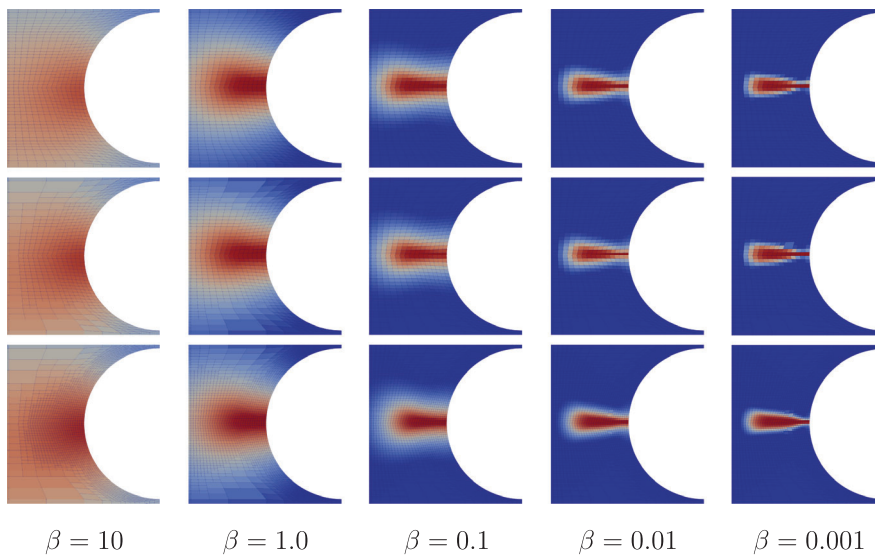


FIGURE 17 Double-notched geometry: loading stepsizes and iteration counts for damage update and elasticity solver for a simulation with a maximal level of refinement $L_{\max} = 3$, loading stepsize bounds $d\hat{u}_{\min} = 0.1 \text{ nm}$ and $d\hat{u}_{\max} = 1 \text{ } \mu\text{m}$, and $\beta = 1.0 \text{ MPa mm}^2$



(A) Comparison of full and adaptive mesh refinement using a regularization $\beta = 1.0 \text{ MPa mm}^2$

(B) Comparison of different regularization parameter β under adaptive mesh refinement



(C) Damage function f at the damage initialization (zoom at right notch) for full refinement for $L_{\max} = 3$ (top), and adaptive refinement with $L_{\max} = 3$ (middle) and 4 (bottom)

FIGURE 18

Double-notched geometry: force-displacement curves for simulations with loading stepsize bounds $d\hat{u}_{\min} = 0.1 \text{ nm}$ and $d\hat{u}_{\max} = 1 \text{ } \mu\text{m}$. All β are stated in MPa mm^2 . A, Comparison of full and adaptive mesh refinement using a regularization $\beta = 1.0 \text{ MPa mm}^2$. B, Comparison of different regularization parameters β under adaptive mesh refinement. C, Damage function f at the damage initialization (zoom at right notch) for full refinement for $L_{\max} = 3$ (top), and adaptive refinement with $L_{\max} = 3$ (middle) and 4 (bottom)

FIGURE 19 Double-notched geometry for $L_{\max} = 3$ and $\beta = 1.0 \text{ MPa mm}^2$: A, A symmetric damage propagation is computed for a symmetric meshing, B, changing the boundary description to a non-symmetric problem by slightly moving a single boundary node of the top notch results in a non-symmetric damage propagation, and C, using a symmetric boundary mesh but changing one interior mesh node produces a non-symmetric damage pattern [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

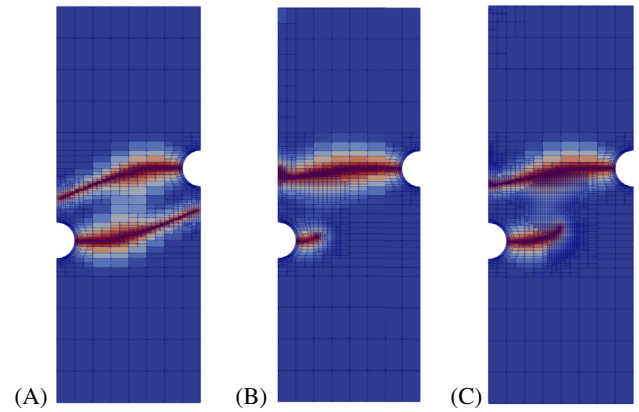
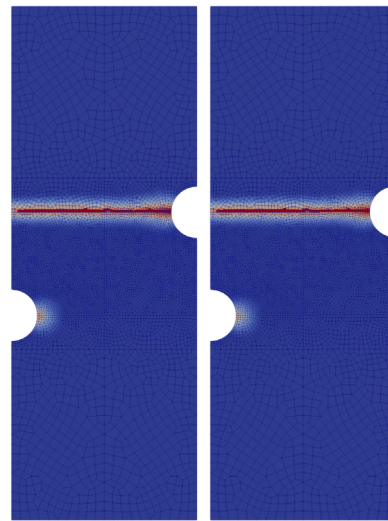
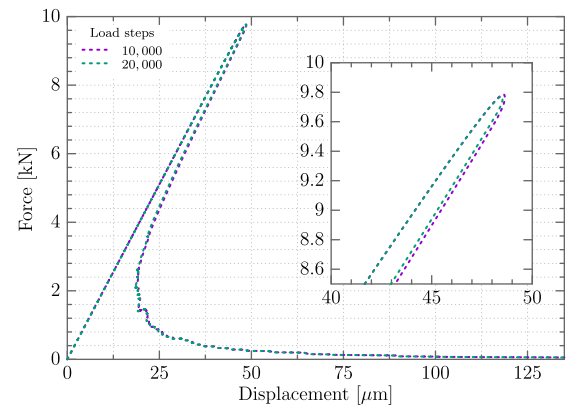


FIGURE 20 Double-notched geometry: Results employing arc length control. Asymmetric evolution of damage is observed due to bifurcation of the solution with, A, 10 000 and, B, 20 000 load steps to resolve the same curve length. C, Force-displacement curves comparing both load step counts showing snapback effects [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]



(A) 10,000 steps (B) 20,000 steps



(C) Force-displacement showing snapback

5 | CONCLUSIONS AND OUTLOOK

We presented a sophisticated study for the convergence properties of a gradient-enhanced brittle material model employing adaptive numerical strategies for load stepping and mesh handling, and analyzed the influence of the numerical parameters on the solution accuracy. We found that adaptive strategies for load stepping and mesh refinement are advantageous in terms of runtime and memory consumption while maintaining the desired degree of accuracy compared to uniformly resolved approaches. This allows us to perform highly accurate simulations, which accurately resolve the force drop in a tensile test, and we thus have a numerical tool at hand, which efficiently and accurately computes the crack load for a given geometry and chosen model parameters. Therefore, we were able to study the influence of the regularization parameter on the solution accuracy without interference with other numerical artifacts and found a very large dependency of the displacement at which crack initiation occurs on the regularization parameter.

The present contribution opens the door for a couple of subsequent works: first, we want to compare our numerical predictivity to experiments. To this end, the dissipation parameter r has to be calibrated for a prescribed regularization parameter β and should be predictive for several geometry types. Second, an a priori fixed maximum level of refinement is not purposeful. Instead, the local mesh resolution should be adapted based on the chosen regularization parameter. A good heuristic for the fracture width may allow for a mesh refinement criterion solely based on β and the local mechanical energy, and thereby render the parameter $L_{\max}$ obsolete. This is part of our ongoing work. Third, the imposed loading step bounds $d\hat{\mathbf{u}}_{\min}$ and $d\hat{\mathbf{u}}_{\max}$ require an educated a priori guess. Ideally, the loading stepsize should solely depend on the required accuracy and therefore only be controlled by the ϵ_{load} parameter or similar error measures. Fourth, the iteration counts for the damage update increase quite drastically under mesh refinement. This is still acceptable due to the very short execution time for one single iterate. However, it must be carefully taken into account for very large geometries or

high resolutions. Avoiding a Jacobi-like update scheme in favor of more elaborate iterative methods seems reasonable here. Alternatively, the operator split may be avoided using a monolithic simulation approach, still allowing for adaptivity in loading and mesh resolution. Finally, the proposed mesh can be parallelized for a further speedup of computation and to allow for simulations of large and complicated geometries.

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REFERENCES

1. Hertzberg RW. *Deformation and Fracture Mechanics of Engineering Materials*. New York: Wiley; 1989.
2. Anderson TL. *Fracture Mechanics: Fundamentals and Applications*. Boca Raton, FL: CRC Press; 2017.
3. Gross D, Seelig T. *Fracture Mechanics: With an Introduction to Micromechanics*. Cham, Switzerland: Springer; 2017.
4. Schmidt B, Fraternali F, Ortiz M. Eigenfracture: an eigendeformation approach to variational fracture. *Multiscale Model Sim*. 2009;7(3):1237-1266.
5. Pandolfi A, Ortiz M. An eigenerosion approach to brittle fracture. *Int J Numer Methods Eng*. 2012;92(8):694-714.
6. Miehe C, Welschinger F, Hofacker M. Thermodynamically consistent phase-field models of fracture: variational principles and multi-field FE implementations. *Int J Numer Methods Eng*. 2010;83(10):1273-1311.
7. Miehe C, Hofacker M, Welschinger F. A phase field model for rate-independent crack propagation: Robust algorithmic implementation based on operator splits. *Comput Methods Appl Mech Eng*. 2010;199(45):2765-2778.
8. Strobl M, Seelig T. A novel treatment of crack boundary conditions in phase field models of fracture. *PAMM*. 2015;15(1):155-156.
9. Strobl M, Seelig T. On constitutive assumptions in phase field approaches to brittle fracture. *Procedia Struct Integrity*. 2016;2:3705-3712.
10. Lemaitre J, Desmorat R. *Engineering Damage Mechanics: Ductile, Creep, Fatigue and Brittle Failures*. Berlin, Heidelberg, Germany: Springer-Verlag; 2005.
11. Wriggers P, Moftah SO. Mesoscale models for concrete: Homogenisation and damage behaviour. *Finite Elem Anal Des*. 2006;42(7):623-636.
12. Matzenmiller ALJTR, Lubliner J, Taylor RL. A constitutive model for anisotropic damage in fiber-composites. *Mech Mater*. 1995;20(2):125-152.
13. Menzel A, Ekh M, Runesson K, Steinmann P. A framework for multiplicative elastoplasticity with kinematic hardening coupled to anisotropic damage. *Int J Plast*. 2005;21(3):397-434.
14. Altenbach H, Huang C, Naumenko K. Creep-damage predictions in thin-walled structures by use of isotropic and anisotropic damage models. *J Strain Anal Eng Des*. 2002;37(3):265-275.
15. Naumenko K, Altenbach H, Kutschke A. A combined model for hardening, softening, and damage processes in advanced heat resistant steels at elevated temperature. *Int J Damage Mech*. 2011;20(4):578-597.
16. Brepols T, Wulfinghoff S, Reese S. Gradient-extended two-surface damage-plasticity: micromorphic formulation and numerical aspects. *Int J Plast*. 2017;97:64-106.
17. Brepols T, Wulfinghoff S, Reese S. A micromorphic damage-plasticity model to counteract mesh dependence in finite element simulations involving material softening. *Multiscale Modeling of Heterogeneous Structures*. Cham, Switzerland: Springer; 2018:235-255.
18. Bazant Z, Pijaudier-Cabot G, Pan J. Ductility, snapback, size effect, and redistribution in softening beams or frames. *J Struct Eng-ASCE*. 1987;113:12.
19. Bažant ZP, Jirásek M. Nonlocal integral formulations of plasticity and damage: survey of progress. *J Eng Mech*. 2002;128(11):1119-1149.
20. Peerlings RHJ, Geers MGD, de Borst R, Brekelmans WAM. A critical comparison of nonlocal and gradient-enhanced softening continua. *Int J Solids Struct*. 2001;38(44):7723-7746.
21. Faria R, Oliver J, Cervera M. A strain-based plastic viscous-damage model for massive concrete structures. *Int J Solids Struct*. 1998;35(14):1533-1558.
22. Needleman A. Material rate dependence and mesh sensitivity in localization problems. *Comput Methods Appl Mech Eng*. 1988;67:69-85.
23. Suffis A, Ton A, Lubrecht A, Combescure A. Damage model with delay effect: analytical and numerical studies of the evolution of the characteristic damage length. *Int J Solids Struct*. 2003;40(13-14):3463-3476.
24. Forest S, Lorentz E. Localization phenomena and regularization methods. In: Besson J, ed. *Local approach to fracture*, Paris: Presses des Mines; vol 1. 2004:311-371.
25. Junker P, Schwarz S, Makowski J, Hackl K. A relaxation-based approach to damage modeling. *Contin Mech Thermodyn*. 2017;29:291-310.
26. Langenfeld K, Junker P, Mosler J. Quasi-brittle damage modeling based on incremental energy relaxation combined with a viscous-type regularization. *Continuum Mechanics and Thermodynamics*; 2018:1-20.
27. de Borst R. Some recent issues in computational failure mechanics. *Int J Numer Methods Eng*. 2001;52(1-2):63-95.
28. Dimitrijevic B, Hackl K. A method for gradient enhancement of continuum damage models. *Tech Mech*. 2008;28(1):43-52.
29. Junker P, Hackl K. A variational growth approach to topology optimization. *Struct Multidiscip Optim*. 2015;52(2):293-304.
30. Sigmund O, Maute K. Topology optimization approaches. *Struct Multidiscip Optim*. 2013;48(6):1031-1055.
31. Labergere C, Rassineux A, Saanouni K. Numerical simulation of continuous damage and fracture in metal-forming processes with 2d mesh adaptive methodology. *Finite Elem Anal Des*. 2014;82:46-61.

32. Junker P, Schwarz S, Dustin DR, Hackl K. A fast and robust numerical treatment of a gradient-enhanced model for brittle damage. *Int J Multiscale Comput Eng*. 2019;17(2):151-180.
33. Coirier WJ. An Adaptively-Refined, Cartesian, Cell-Based Scheme for the Euler and Navier-Stokes Equations [Ph.D. thesis]. The University of Michigan; also published as NASA TM 106754; 1994.
34. Sadat H, Prax C. Application of the diffuse approximation for solving fluid flow and heat transfer problems. *Int J Heat Mass Transf*. 1996;39(1):214-218.
35. Hackbusch W. *Multi-Grid Methods and Applications*. Vol 4. Berlin, Heidelberg and New York: Springer; 1985.
36. Reiter S, Vogel A, Heppner I, Rupp M, Wittum G. A massively parallel geometric multigrid solver on hierarchically distributed grids. *Comput Vis Sci*. 2013;16(4):151-164.
37. Bramble JH, Pasciak JE, Wang JP, Xu J. Convergence estimates for multigrid algorithms without regularity assumptions. *Math Comput*. 1991;57(195):23-45.
38. Bastian P. *Parallele Adaptive Mehrgitterverfahren*. Stuttgart, Germany: Teubner; 1996.
39. Vogel A, Reiter S, Rupp M, Nägel A, Wittum G. UG 4: a novel flexible software system for simulating PDE based models on high performance computers. *Comput Vis Sci*. 2013;16(4):165-179.
40. Promesh. www.promesh3d.com.
41. Paraview. www.paraview.org.
42. Finite element analysis program. <http://projects.ce.berkeley.edu/feap/>.

SUPPORTING INFORMATION

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